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Apparatus and method for enlarging an incision

Abstract

A retraction system and method are provided for retracting tissues surrounding a surgical site. In one aspect, a method including engaging slide connections between a guide dilator and a plurality of tissue engaging members and sequentially enlarging an incision using the guide dilator and the plurality of tissue engaging members. In another aspect, a method of inserting a plurality of tissue engaging members into an incision including fixing tip portions of the plurality of tissue engaging members in an insertion configuration, advancing the tip portions into an incision, and restricting movement of the tip portions away from the insertion configuration. A guide dilator system comprising an elongate body, a plurality of tissue engaging members, and slide connections between the elongate body and the tissue engaging members is also provided.

Inventors: Perrow; Scott J. (Ishpeming, MI), Riceman; Bruce Anthony (Leander, TX), Colantonio; Jacob Anton (Marquette, MI)

Applicant: Xtant Medical Holdings, Inc. (Belgrade, MT)

Family ID: 63581353

Assignee: XTANT MEDICAL HOLDINGS, INC. (Belgrade, MT)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2473266	12/1948	Wexler	N/A	N/A
2623517	12/1951	Israel et al.	N/A	N/A
3038468	12/1961	Raeuchle	N/A	N/A
3724449	12/1972	Gauthier	N/A	N/A
3948259	12/1975	Bolduc et al.	N/A	N/A
3998217	12/1975	Trumbull et al.	N/A	N/A
4010741	12/1976	Gauthier	N/A	N/A
4434791	12/1983	Darnell	N/A	N/A
5616117	12/1996	Dinkler et al.	N/A	N/A
5667481	12/1996	Villalta et al.	N/A	N/A
5688223	12/1996	Rosendahl	N/A	N/A
5728046	12/1997	Mayer et al.	N/A	N/A
5813978	12/1997	Jako et al.	N/A	N/A
5928139	12/1998	Koros et al.	N/A	N/A
5944658	12/1998	Koros et al.	N/A	N/A
5954635	12/1998	Foley et al.	N/A	N/A
5957927	12/1998	Magee et al.	N/A	N/A

5967973	12/1998	Sherts et al.	N/A	N/A
6083154	12/1999	Liu et al.	N/A	N/A
6139493	12/1999	Koros et al.	N/A	N/A
6206826	12/2000	Mathews et al.	N/A	N/A
6213940	12/2000	Sherts et al.	N/A	N/A
6224545	12/2000	Cocchia et al.	N/A	N/A
6322500	12/2000	Sikora et al.	N/A	N/A
6416467	12/2001	McMillin et al.	N/A	N/A
6431025	12/2001	Koros et al.	N/A	N/A
6464634	12/2001	Fraser	N/A	N/A
6616605	12/2002	Wright et al.	N/A	N/A
6800084	12/2003	Davison et al.	N/A	N/A
6849064	12/2004	Hamada	N/A	N/A
6869398	12/2004	Obenchain et al.	N/A	N/A
6896654	12/2004	Paolitto et al.	N/A	N/A
6932764	12/2004	Kashyap	N/A	N/A
7195592	12/2006	Ravikumar et al.	N/A	N/A
7780594	12/2009	Hutton	N/A	N/A
7931589	12/2010	Cohen et al.	N/A	N/A
7985179	12/2010	Gephart et al.	N/A	N/A
8550995	12/2012	Frasier et al.	N/A	N/A
8702600	12/2013	Perrow	N/A	N/A
8882661	12/2013	Hutton et al.	N/A	N/A
8956285	12/2014	Gephart et al.	N/A	N/A
9113852	12/2014	Perrow	N/A	N/A
9498200	12/2015	Pfabe et al.	N/A	N/A
9579095	12/2016	Perrow	N/A	N/A
9980714	12/2017	Perrow	N/A	N/A
10285680	12/2018	Friedrich et al.	N/A	N/A
10357239	12/2018	Perrow et al.	N/A	N/A
11172919	12/2020	Perrow	N/A	A61B 17/0293
11980356	12/2023	Perrow et al.	N/A	N/A
2002/0111538	12/2001	Wright et al.	N/A	N/A
2002/0193666	12/2001	Sherts et al.	N/A	N/A
2003/0191371	12/2002	Smith et al.	N/A	N/A
2004/0002629	12/2003	Branch et al.	N/A	N/A
2004/0059193	12/2003	Fanous	N/A	N/A
2004/0087833	12/2003	Bauer et al.	N/A	N/A
2004/0133077	12/2003	Obenchain et al.	N/A	N/A
2004/0133201	12/2003	Shluzas et al.	N/A	N/A
2004/0176665	12/2003	Branch et al.	N/A	N/A
2004/0193018	12/2003	Thalgott et al.	N/A	N/A
2004/0230100	12/2003	Shluzas	N/A	N/A
2004/0242969	12/2003	Sherts et al.	N/A	N/A
2005/0080320	12/2004	Lee et al.	N/A	N/A
2005/0137461	12/2004	Marchek et al.	N/A	N/A
2005/0149035	12/2004	Pimenta et al.	N/A	N/A
2005/0159650	12/2004	Raymond et al.	N/A	N/A
2005/0159651	12/2004	Raymond et al.	N/A	N/A

2005/0192485	12/2004	Branch et al.	N/A	N/A
2005/0215862	12/2004	Larson et al.	N/A	N/A
2005/0215866	12/2004	Kim	N/A	N/A
2005/0234304	12/2004	Dewey et al.	N/A	N/A
2005/0261694	12/2004	Orton et al.	N/A	N/A
2005/0277812	12/2004	Myles	N/A	N/A
2006/0004261	12/2005	Douglas	N/A	N/A
2006/0030858	12/2005	Simonson et al.	N/A	N/A
2006/0052672	12/2005	Landry et al.	N/A	N/A
2006/0074445	12/2005	Gerber et al.	N/A	N/A
2006/0224044	12/2005	Marchek et al.	N/A	N/A
2007/0010716	12/2006	Malandain et al.	N/A	N/A
2007/0118022	12/2006	Hutton	N/A	N/A
2007/0156025	12/2006	Marchek et al.	N/A	N/A
2007/0161867	12/2006	Fowler, Jr. et al.	N/A	N/A
2007/0238932	12/2006	Jones et al.	N/A	N/A
2007/0282171	12/2006	Karpowicz et al.	N/A	N/A
2008/0033251	12/2007	Araghi	N/A	N/A
2008/0188718	12/2007	Spitler	600/232	A61B 17/0206
2008/0214898	12/2007	Warren	N/A	N/A
2009/0069635	12/2008	Gephart et al.	N/A	N/A
2015/0209022	12/2014	Ruppert et al.	N/A	N/A
2016/0174997	12/2015	Spitznagel	N/A	N/A
2016/0331431	12/2015	Gephart	N/A	N/A
2018/0029824	12/2017	Gephart et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 03/086202	12/2002	WO	N/A
WO 2004/047650	12/2003	WO	N/A
WO 2005/092206	12/2004	WO	N/A
WO 2005/094695	12/2004	WO	N/A
WO 2005/096735	12/2004	WO	N/A
WO 2007/087536	12/2006	WO	N/A
WO 2008/039427	12/2007	WO	N/A
WO 2014/140100	12/2013	WO	N/A

OTHER PUBLICATIONS

Clarity Lateral, Retractor System, Surgical Technique, 2008, 12 pages. cited by applicant

ProAccess Radiolucent Retractor Blades, Sythes Spine, product offerings guide provided by the manufacturer, 2004, 2 pages, Paoli, Pennsylvania. cited by applicant

SynFrame Access and Retractor System Assembly Guide, Synthes Spine, assembly guide provided by the manufacturer, 1999, 12 pages, Paoli, Pennsylvania. cited by applicant

Webb, J., Spine-the future, AO foundation Webpage, available at:
http://www.aofoundation.org/AOFileServer/PortalFiles?FilePath=/Extranet2007/active/_att/wor/act/Dialogue/1999_2/spine.pdf, accessed Apr. 10, 2009.
cited by applicant

TeDan Surgical Innovations, Phantom XL Series brochure, copyright dale 2013, 2 pages. cited by applicant

Official Action for U.S. Appl. No. 13/415,673, dated Feb. 4, 2013, 10 pages. cited by applicant
Official Action for U.S. Appl. No. 13/415,673, dated Sep. 16, 2013, 9 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 13/415,673, dated Dec. 5, 2013, 7 pages. cited by applicant
Official Action for U.S. Appl. No. 14/250,063, dated Jul. 1, 2014, 10 pages. cited by applicant
Official Action for U.S. Appl. No. 14/250,063, dated Jan. 28, 2015, 8 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 14/250,063, dated Apr. 20, 2015, 5 pages. cited by applicant
Official Action for U.S. Appl. No. 14/832,592, dated Oct. 30, 2015, 6 pages. cited by applicant
Official Action for U.S. Appl. No. 14/832,592, dated May 9, 2016, 8 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 14/832,592, dated Aug. 11, 2016, 12 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 14/832,592, dated Jan. 24, 2017, 3 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 15/382,192, dated Jan. 26, 2018, 11 pages. cited by applicant
Official Action for U.S. Appl. No. 15/991,502, dated Aug. 7, 2018, 8 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 15/991,502, dated Mar. 11, 2019, 5 pages. cited by applicant
Official Action for U.S. Appl. No. 16/458,945, dated Mar. 23, 2021, 8 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 16/458,945, dated Jul. 15, 2021, 7 pages. cited by applicant
Notice of Allowance for U.S. Appl. No. 17/526,766, dated Dec. 22, 2023, 12 pages. cited by applicant
U.S. Appl. No. 13/415,673, filed Mar. 8, 2012, now issued as U.S. Pat. No. 8,702,600. cited by applicant
U.S. Appl. No. 14/250,063, filed Apr. 10, 2014, now issued as U.S. Pat. No. 9,113,852. cited by applicant
U.S. Appl. No. 14/832,592, filed Aug. 21, 2015, now issued as U.S. Pat. No. 9,579,095. cited by applicant
U.S. Appl. No. 15/382,192, filed Dec. 16, 2016, now issued as U.S. Pat. No. 9,980,714. cited by applicant
U.S. Appl. No. 15/991,502, filed May 29, 2018, now issued as U.S. Pat. No. 10,357,239 cited by applicant
U.S. Appl. No. 16/458,945, filed Jul. 1, 2019, now issued as U.S. Pat. No. 11,172,919. cited by applicant
U.S. Appl. No. 17/526,766, filed Nov. 15, 2021, now issued as U.S. Pat. No. 11,980,356. cited by applicant

Primary Examiner: Robert; Eduardo C

Assistant Examiner: Negrelli-rodriguez; Christina

Attorney, Agent or Firm: Sheridan Ross P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/526,766, filed Nov. 15, 2021, now U.S. Pat. No. 11,980,356, which is a continuation of U.S. patent application Ser. No. 16/458,945, filed Jul. 1, 2019, now U.S. Pat. No. 11,172,919, which is a continuation of U.S. patent application Ser. No. 15/991,502, filed

May 29, 2018, which issued as U.S. Pat. No. 10,357,239 on Jul. 23, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 15/382,192, filed Dec. 16, 2016, which issued as U.S. Pat. No. 9,980,714 on May 29, 2018, which is a continuation of U.S. patent application Ser. No. 14/832,592, filed Aug. 21, 2015, which issued as U.S. Pat. No. 9,579,095 on Feb. 28, 2017, which is a continuation-in-part of U.S. patent application Ser. No. 14/250,063, filed Apr. 10, 2014, which issued as U.S. Pat. No. 9,113,852 on Aug. 25, 2015, which is a continuation of U.S. patent application Ser. No. 13/415,673, filed Mar. 8, 2012, which issued as U.S. Pat. No. 8,702,600 on Apr. 22, 2014, which claims the benefit of U.S. Provisional Patent Application No. 61/450,560, filed Mar. 8, 2011, all of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

(1) The invention relates to a system and method for providing access to a surgical site and, more particularly, to a system and method for retracting tissue during a surgical procedure.

BACKGROUND OF THE INVENTION

(2) Advancing blades of a surgical retractor through an incision and into position adjacent a surgical site involves overcoming resistance from muscles, tendons, and other tissues within the body. When the surgical retractor is used to access a patient's spine from the side of the patient, known as lateral approach surgery, the distance to the patient's spine is greater than if the spine were accessed through the patient's back, known as posterior approach surgery. The increased distance to the spine during lateral approach surgery means overcoming a greater amount of resistance from muscles, tendons, and other tissues in order to advance the retractor blades to the surgical site than in a posterior approach surgery.

(3) Another difficulty with lateral approach surgery is that the tissues surrounding the retractor blades tightly envelop and deflect the blades as the blades are slid along a dilator toward the surgical site. The increased distance to the spine for lateral approach surgery means that the retractor blades must be longer in order to retract tissues adjacent the surgical site. Like cantilever beams, the longer retractor blades used for lateral approach surgery are subjected to forces from the surrounding tissues that deflect the blades a greater amount than if the blades were shorter blades typically used for posterior approach surgery. Deflection of the long retractor blades allows the tissues enveloping the blades to lodge between the blades and the dilator as well as creep into gaps between the blades. The retractor blades may have to be removed and re-inserted into the patient to remove tissues caught between the blades and the dilator or caught in gaps between the blades, which complicates the surgical procedure.

(4) For longer length blades, such as the 80 mm-180 mm blades typically used for lateral approach surgery, small manufacturing variations in straightness along the blades are magnified by the length of the blades. These variations are problematic, as even a slight curvature may cause the tip of a longer blade to lift up from the dilator and trap tissue as the blade is advanced toward the surgical site. Variations in blade straightness may also cause the tips of the retractor blades to be unevenly spaced around the dilator and permit tissue creep between the retractor blades. These situations increase the difficulty of advancing the retractor blades toward the surgical site and complicate the process of establishing a working channel to the spine.

SUMMARY OF THE INVENTION

(5) In one aspect of the invention, a method is provided for enlarging an incision that limits tissue creep as tissue engaging members are advanced into an incision and substantially improves the ease with which a surgeon may establish a working channel within a patient. More specifically, the method involves engaging slide connections between a plurality of tissue engaging members and a guide dilator which restricts separation of the tissue engaging members and accompanying tissue creep therebetween. Engaging slide connections between the tissue engaging members and the guide dilator also limits tissues lodging between the tissue engaging members and the guide dilator by resisting separation of the tissue engaging members from the guide dilator. Further, the method

involves advancing a leading end portion of the guide dilator into the incision to enlarge the incision to an initial opening and advancing distal end portions of the tissue engaging members connected to the guide dilator into the incision to enlarge the initial opening to a larger, intermediate-sized opening. In this manner, the leading end portion of the guide dilator and the distal end portions of the tissue engaging members sequentially dilate the incision and minimize strain on tissues surrounding the incision. In one approach, the method further comprises withdrawing the leading end portion of the guide dilator from the incision while restricting movement of the plurality of the tissue engaging members. This disengages the slide connections between the plurality of tissue engaging members and the guide dilator without having to individually disconnect the tissue engaging members or utilize another tool to disconnect the tissue engaging members from the guide dilator. Accordingly, the present method provides an easier and faster approach for advancing tissue engaging members into position adjacent a surgical site while limiting interference from bodily tissues.

(6) In another aspect of the invention, a method is provided for inserting tissue engaging members into an incision that limits tissue creep even if the tissue engaging members are longer and have varying degrees of straightness. The method includes fixing tip portions of the tissue engaging members in a predetermined insertion configuration and advancing the tip portions into an incision. This way, any variation in the straightness of one or more of the tissue engaging members is compensated for by manipulating the associated tip portion(s) into the predetermined insertion configuration before the tip portions are advanced into the incision. The method further comprises restricting movement of the tip portions of the tissue engaging members away from the insertion configuration. In this manner, the tip portions may be held in the insertion configuration against resistance from bodily tissues as the tip portions are advanced within the patient. In one approach, fixing the tip portions of the plurality of tissue engaging members comprises engaging slide connections between the tissue engaging members and a guide dilator. The slide connections permit a surgeon to rapidly connect the tissue engaging members to the guide dilator and insert the guide dilator and tissue engaging members into the incision without muscle, tendons, or other tissues shifting the tip portions of the tissue engaging members from the predetermined insertion configuration.

(7) A guide dilator system is also provided including an elongate body, a plurality of tissue engaging members, and slide connections between the elongate body and the plurality of tissue engaging members. In this manner, the tissue engaging members can be easily engaged with the elongate body, advanced into an incision, and disengaged from the elongate body adjacent a surgical site. In one form, the slide connections comprise axially extending surfaces of each slide connection that extend a majority of a predetermined length of the respective tissue engaging members with the tissue engaging members in an operative position. In this manner, the slide connections provide a firm engagement between the tissue engaging members and the elongate body when the tissue engaging members are in an operative position on and connected to the elongate body.

(8) The elongate body and the tissue engaging members may further include stop portions configured to abut and limit sliding of the tissue engaging members along the elongate body in a predetermined direction. For example, the predetermined direction can be from a leading end portion of the elongate body toward a trailing end portion of the elongate body. As the leading end portion of the elongate body is advanced toward the surgical site, the stop portions of the elongate body and the tissue engaging members abut and keep the tissue engaging members from sliding off of the elongate body as the surrounding tissues resist movement of the elongate body and the tissue engaging members within the patient. Once the tissue engaging members are positioned adjacent a surgical site, the elongate body is withdrawn from the incision which slides the tissue engaging members off of the elongate body. In this manner, the guide dilator system can simplify and

accelerate the process of advancing the tissue engaging members into position adjacent a surgical site.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a guide dilator and retractor blades in accordance with one form of the present invention showing one of the retractor blades slid off of the guide dilator;
- (2) FIG. 2 is an enlarged, end view of the guide dilator of FIG. 1 with all of the retractor blades connected to the guide dilator showing an outer profile of the retractor blades;
- (3) FIG. 3 is a perspective view of the guide dilator of FIG. 1 showing mounting rails of the guide dilator for slidably engaging in slots of the retractor blades of FIG. 1;
- (4) FIG. 3A is a cross sectional view of the guide dilator taken across line 3A-3A in FIG. 3 showing flanges of the mounting rails which engage cooperating slots on the retractor blades;
- (5) FIG. 4 is an enlarged, side elevational view of the mounting rails of the guide dilator of FIG. 1 showing resilient members of the mounting rails;
- (6) FIG. 5 is a perspective view of one of the retractor blades of FIG. 1 showing a slot of the blade for slidably receiving a mounting rail of the guide dilator and a cannula disposed at the distal end of the retractor blade;
- (7) FIG. 5A is a cross-sectional view taken across the line 5A-5A in FIG. 5A
- (8) FIG. 6 is a perspective view of another one of the retractor blades of FIG. 1 showing a slot of the blade for slidably receiving a mounting rail of the guide dilator, the retractor blade of FIG. 6 lacking a cannula similar to the retractor blade of FIG. 5;
- (9) FIG. 7 is an enlarged, perspective view of a leading end of a guide dilator in accordance with another form of the present invention showing mounting rails of the guide dilator;
- (10) FIG. 7A is a cross sectional view taken across line 7A-7A in FIG. 7 showing a resilient member of one of the mounting rails of the guide dilator of FIG. 7;
- (11) FIG. 7B is a cross sectional view similar to FIG. 7A showing the resilient member deflected downward and engaged with a retractor blade connected to the mounting rail;
- (12) FIG. 8 is a perspective view of a guide dilator and retractor blades in accordance with another form of the present invention showing one of the retractor blades slid off of the guide dilator;
- (13) FIG. 9 is an enlarged, end view of the guide dilator of FIG. 8 with both retractor blades connected to the guide dilator showing an outer profile of the retractor blades;
- (14) FIG. 10 is a perspective view of one of the retractor blades of FIG. 8 showing a slot of the blade for slidably receiving a mounting rail of the guide dilator;
- (15) FIGS. 11-24 illustrate an exemplary method of enlarging an incision during a lateral approach surgery using the guide dilator and retractor blades of FIG. 8;
- (16) FIG. 25 is a perspective view of a retractor in accordance with one form of the present invention showing one of the sliders of the retractor in a retracted, pivoted position;
- (17) FIG. 26 is a perspective view of a wedge lock of a slider of the retractor of FIG. 25 showing the wedge lock removed from the slider;
- (18) FIG. 27 is a perspective view of a slider inner member of the retractor of FIG. 25 showing the inner member removed from the retractor;
- (19) FIG. 28 is an enlarged, perspective view of a slider and portions of a ratchet mechanism of the retractor of FIG. 25 showing the slider and the ratchet mechanism portions removed from the retractor;
- (20) FIG. 29 is an enlarged, perspective view of a slider of the retractor of FIG. 25 showing an inner member of the slider having its rear end portion pivoted upward;
- (21) FIG. 30 is a perspective view of a retractor in accordance with another form of the present

invention showing retractor blades connected to sliders which are retracted and pivoted;

(22) FIG. **31** is an elevational view of a retractor blade docking anchor showing a pointed tip of the docking anchor for penetrating an intervertebral disc;

(23) FIG. **32** is a perspective view of a light source showing a flexible fiber optic cable;

(24) FIG. **33** is a perspective view of a retractor in accordance with another form of the present invention showing an elevated portion of the retractor frame;

(25) FIG. **34** is a perspective view of the retractor of FIG. **33** showing blades mounted on the retractor and disposed adjacent an intervertebral disc between L4 and L5 vertebrae;

(26) FIG. **35** is a perspective view of a pin dilator that may be used with the guide dilators of FIGS. **1**, **7**, and **8** in place of a separate guide wire and a first dilator;

(27) FIG. **36** is a perspective view of a retractor in accordance with another form of the present invention that includes locking mechanisms for locking sliders of the retractor;

(28) FIG. **37** is a perspective view of the retractor of FIG. **36** showing sliders and attached retractor blades of the retractor in a retracted configuration which generally forms a working channel between the blades;

(29) FIG. **38** is a perspective view of the retractor of FIG. **36** showing one of the sliders in a beyond center position where the retractor blade connected to the slider has been moved radially inward to a point beyond a center of the working channel;

(30) FIG. **39** is a perspective view of the slider shown in the beyond center position in FIG. **38** showing the slider removed from the retractor frame and a dashed circle extending about a locking mechanism of the slider;

(31) FIG. **40** is a cross sectional view taken across line **40-40** in FIG. **37** showing the slider in a retracted position and a lock button of the locking mechanism in a lowered locked position;

(32) FIG. **41** is a cross sectional view similar to FIG. **40** showing the slider in the beyond center position and the lock button in a raised unlocked position;

(33) FIG. **42** is a perspective view of a retractor in accordance with another form of the present invention showing sliders of the retractor in extended positions thereof;

(34) FIG. **43** is a perspective view of one of the sliders of the retractor of FIG. **42** showing a slider linear drive and a ratchet mechanism engaged on opposite sides of the slider;

(35) FIG. **44** is a cross-sectional view taken across line **44-44** in FIG. **42** showing a drive cap and a pinion of the slider linear drive received in a bore hole of the frame;

(36) FIG. **45** is a top perspective view of the pinion of FIG. **44** showing an upper portion of the pinion having inclined cam surfaces and vertically extending drive surfaces;

(37) FIG. **46** is a bottom perspective view of the drive cap of FIG. **44** showing inclined cam surfaces and vertically extending drive surfaces;

(38) FIG. **47** is a perspective view of a button and a lock latch of the ratchet mechanism of FIG. **43** showing a tooth of the button for engaging the slider;

(39) FIG. **47A** is a perspective view of the lock latch of FIG. **47** and an internal detent mechanism of the button showing a ball of the detent mechanism engaging a recess of the lock latch to resist movement of the lock latch away from a selected orientation thereof;

(40) FIG. **48** is a cross-sectional view taken across line **48-48** in FIG. **42** showing the lock latch in an unlocked position and the tooth of the button disengaged from the slider;

(41) FIG. **49** is a top plan view of the portion of the retractor of FIG. **48** showing the lock latch received in a cavity of the frame and abutting a wall of the cavity;

(42) FIG. **50** is a cross-sectional view similar to FIG. **48** showing the lock latch pivoted to a ratchet position and the tooth of the button engaged with the slider;

(43) FIG. **51** is a top plan view of the portion of the retractor of FIG. **50** showing the lock latch in the ratchet orientation and the lock latch having clearance within the frame cavity to permit the lock latch to pivot back and forth with the button as the tooth thereof ratchets in and out of depressions of the slider;

- (44) FIG. 52 is a cross-sectional view similar to FIG. 48 showing the lock latch pivoted to a locked position such that the lock latch abuts a surface of the frame and the tooth of the button is fixed in engagement in a depression of the slider;
- (45) FIG. 53 is a top plan view of the portion of the retractor of FIG. 52 showing the lock latch holding the button and tooth thereof in fixed engagement with the slider;
- (46) FIG. 54 is a cross-sectional view taken across line 54-54 in FIG. 43 showing an inner member of the slider in an initial configuration;
- (47) FIG. 55 is a cross-sectional view similar to FIG. 54 showing an elevation screw of the slider having been rotated to pivot the inner member to an inclined orientation relative to an outer member of the slider;
- (48) FIG. 56 is a perspective view of a retractor having operating mechanisms that receive tissue engaging members and permit retraction of the tissue engaging members to retract tissues;
- (49) FIG. 57 is a perspective view of a slider and a slider linear drive of one of the operating mechanisms of FIG. 56;
- (50) FIG. 58 is a perspective view of a worm screw and a worm gear assembly of the linear drive mechanism of FIG. 57;
- (51) FIG. 59 is a cross-sectional view taken across line 59-59 in FIG. 58 showing a helical tooth of the worm screw engaged with teeth of the worm gear assembly;
- (52) FIG. 60 is an exploded, perspective view of the worm gear assembly of FIG. 58 showing ball bearings that are radially shiftable to couple and decouple a worm gear and a pinion gear of the worm gear assembly;
- (53) FIG. 61A is a cross-sectional view of the worm gear assembly of FIG. 60 in an assembled configuration, FIG. 61A showing a release shaft of the worm gear assembly in a drive position wherein a head portion of the release shaft keeps the ball bearings shifted radially outward and couples the worm gear and the pinion gear;
- (54) FIG. 61B is a cross-sectional view similar to FIG. 61A showing the release shaft in a release position wherein a neck portion of the release shaft permits the ball bearings to shift radially inward and decouple the worm gear and the pinion gear;
- (55) FIG. 62A is a cross-sectional view taken across line 62A-62A in FIG. 61A showing the release shaft in the drive position and the ball bearings extending in through openings of the pinion gear and into pockets of the worm gear;
- (56) FIG. 62B is a cross-sectional view similar to FIG. 61A showing the release shaft in the release position and the ball bearings shifted out of the pockets of the worm gear;
- (57) FIG. 63 is a perspective view of a retractor having operating mechanisms that support tissue engaging members and permit a user to retract the tissue engaging members and retract tissues of a patient;
- (58) FIG. 64 is a perspective view of a slider and a slider linear drive of one of the operating mechanisms of the retractor of FIG. 63;
- (59) FIG. 65 is a cross-sectional view taken across line 65-65 in FIG. 64 showing a wall of an outer member of the slider received in a pocket of an inner member of the slider; and
- (60) FIG. 66 is a perspective view of the outer member of the slider of FIG. 65 showing a through opening of the wall that receives a pivot pin to connect the outer member to the inner member of the slider.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(61) In FIGS. 1-6, a guide dilator 10 for inserting tissue engaging members, such as retractor blades 12, 14, 16, 18, through an incision and into position adjacent a surgical site is provided. The guide dilator 10 provides a rigid support for the retractor blades 12, 14, 16, 18 as the guide dilator 10, and the retractor blades 12, 14, 16, 18 connected thereto, slide along one or more initial dilators 516, 520 (see FIG. 12) into position adjacent a surgical site. The guide dilator 10 supports the retractor blades 12, 14, 16, 18 against resistance from muscles, tendons, and other tissues and to limit

splaying of the retractor blades **12, 14, 16, 18** as they are advanced along the guide dilator **10** into the incision. In this manner, the guide dilator **10** allows a surgeon to easily advance the retractor blades **12, 14, 16, 18** into an incision without the retractor blades **12, 14, 16, 18** shifting and permitting tissues to encroach into gaps between the blades **12, 14, 16, 18**.

(62) With reference to FIG. **1**, the guide dilator **10** has a generally tubular configuration extending along a longitudinal axis **19**. Further, the guide dilator **10** has slide connections with the retractor blades **12, 14, 16, 18**, such as a mounting rail **20** that is received in sliding engagement in a slot **190** (see FIG. **5**) on the blade **12**, which permit the blades **12, 14, 16, 18** to be easily slid onto the guide dilator **10** in direction **22** and off of the guide dilator **10** in direction **44**, as shown in FIG. **1**. This way, the guide dilator **10** and blades **12, 14, 16, 18** can be rapidly and securely assembled during a surgery. The slide connections have stop portions, such as a lower end wall **170** of the mounting rail **20** and a stop surface **174** (see FIG. **5**) of the blade **12**, which are configured to abut and keep the blades **12, 14, 16, 18** from sliding off of the guide dilator **10** in direction **22** as tissues resist movement of the blades **12, 14, 16, 18** into an incision.

(63) In one form, the guide dilator **10** has a tissue engaging member retention mechanism, such as resilient arms **30, 32** of the mounting rail **20** which have distal ends **30a, 30b** configured to be spaced apart by a distance that is greater than a width of the slot **190** of retractor blade **12** (see FIG. **5**). When the blade **12** is slid onto mounting rail **20**, the blade **12** deflects the distal ends **30a, 30b** of the arms **30, 32** toward each other. The deflected arms **30, 32** resiliently bias against and frictionally engage surfaces **193, 195** of the slot **190**. In this manner the tissue engaging member retention mechanisms engage the blades **12, 14, 16, 18** and provide resistance to removal of the blades **12, 14, 16, 18** from the guide dilator **10**, as shown in FIG. **1**. The frictional resistance is sufficient to keep the blades **12, 14, 16, 18** engaged with the guide dilator **10** when the guide dilator **10** is assembled and handled within the operating room, but permit the blades **12, 14, 16, 18** to be slid off of the guide dilator **10** when the blades **12, 14, 16, 18** are at a desired position within a patient. In this manner, the guide dilator **10** provides an elegant solution to the problem of handling the retractor blades **12, 14, 16, 18** throughout a surgical procedure.

(64) A trailing end portion **40** of the guide dilator **10** is grasped after the retractor blades **12, 14, 16, 18** are mounted on the guide dilator **10** and a leading end portion **42** of the guide dilator **10** is advanced into an incision in direction **44**, as shown in FIG. **1**. Once the leading end portion **42** reaches the surgical site through the incision, the blades **12, 14, 16, 18** are slid downward in direction **44** along an outer bearing surface **46** of the leading end portion **42** until distal ends of the retractor blades **12, 14, 16, 18** reach an end **48** of the guide dilator **10**. Next, a surgical retractor is connected to the blades **12, 14, 16, 18** to hold the retractor blades stationary in the incision and the guide dilator **10** is withdrawn from the incision in direction **22**. The relative movement between the guide dilator **10** and the retractor blades **12, 14, 16, 18** withdraws guide dilator mounting rails **20, 92, 94, 96** (see FIG. **3**) from associated slots in the blades **12, 14, 16, 18** and disengages the guide dilator **10** from the blades **12, 14, 16, 18**. In this manner, moving the guide dilator **10** in a linear motion in direction **44** into an incision advances the connected retractor blades **12, 14, 16, 18** to a desired position adjacent a surgical site, and a reverse, linear motion in direction **22** disengages the guide dilator **10** from the blades **12, 14, 16, 18** and withdraws the guide dilator **10** from the incision in a single linear motion. As is apparent, this is an intuitive and speedy procedure for inserting retractor blades into position adjacent a surgical site.

(65) With reference to the blade **12** in FIG. **1**, the connection between the retractor blade **12** and the mounting rail **20** holds a tip **34** of the blade **12** against sliding in rotary direction **36** around the guide dilator **10** and radially away from the guide dilator **10** in direction **38**. The connections between the mounting rails **92, 94, 96** and the blades **14, 16, 18** similarly limit movement of tips **35, 37, 39** of the blades **14, 16, 18** around and radially away from the guide dilator **10**. In contrast to some previous techniques, where tissues could shift retractor blades as the blades were slid along a dilator into an incision, the guide dilator **10** holds retractor blade tips **34, 35, 37, 39** in a

predetermined orientation around the guide dilator **10** against deflection and tissue creep until the guide dilator **10** is disengaged from the retractor blades **12, 14, 16, 18**.

(66) Turning to FIG. 2, the guide dilator **10** has a central cannula **50** that permits the guide dilator **10** and connected retractor blades **12, 14, 16, 18** to be slid over one or more dilators used to initially enlarge an incision, as will be discussed in greater detail below. The blades **12, 14, 16, 18** form a round outer profile **52** around the guide dilator **10** interrupted by gap spacings **54** between adjacent retractor blades. With the blades **12, 14, 16, 18** connected to guide dilator **10** and the blade distal ends **34, 35, 37, 39** spaced from the guide dilator end **48**, as shown in FIG. 1, the generally cylindrical leading end portion **42** of the guide dilator **10** and the round outer profile **52** of the blade distal ends **34, 35, 37, 39** sequentially dilate an incision and minimize stresses on the tissues surrounding the incision. In an alternative embodiment, the guide dilator **10** may lack a central cannula **50** and instead have a generally cylindrical configuration. In this approach, the leading end portion **42** may have a configuration to provide a controlled dilation of the incision.

(67) The guide dilator **10** and connected retractor blades **12, 14, 16, and 18** provide additional functionality once the guide dilator **10** reaches the surgical site. For example, the retractor blades **12, 16** are cannulated blades having cylindrical portions **64, 66** extending radially inward into grooves **70, 72** which extend along the length of the guide dilator **10**, as shown in FIGS. 2, 3, and 5. The cylindrical portions **64, 66** have throughbores or cannulas **74, 76** that are unobstructed even when the blades **12, 16** are connected to the mounting rails **20, 94**. A fixation pin, nerve monitoring probe, or other tool may be inserted along the grooves **70, 72** and into either one of the cannulas **74, 76** once the guide dilator and retractor blades **12** have been advanced to the surgical site. The presence of the unobstructed cannulas **74, 76** also provides a visual window for a surgeon to observe the tissues and bones directly beneath the blades **12, 16** without having to remove the retractor blades **12, 16** from the guide dilator **10**. In one approach, a surgeon can visually check whether one of the cannulas **74, 76** is above a vertebra by looking through the cannula **74, 76** before driving a fixation pin through the cannula **74, 76** and into the vertebra to secure the associated blade **12, 16** to the vertebra.

(68) The mounting rails **20, 92, 94, 96** have features for rapidly engaging and disengaging the retractor blades **12**, as shown in FIGS. 3 and 3A. For example, the mounting rail **20** has a lower base **110** and an upper flange **112**, the flange **112** having two elongate sections **114, 116** separated by the elongate groove **70** therebetween. The elongate sections **114, 116** have integral resilient arms **30, 32** that frictionally engage inner slot surfaces **193, 195** (see FIG. 5) of the retractor blade **12** to resist removal of the blade **12** from the guide dilator **10**. The guide dilator **10** also has mounting rails **92, 96** that are substantially similar to the mounting rails **20, 92** with the exception that the mounting rails **92, 96** are not intersected or divided by the grooves **70, 72**. Instead, the mounting rail **92** has a lower base **120** and an upper flange **122** with resilient arms **134, 136** substantially similar to the arms **30, 32**. Surrounding the lower base **120** of the mounting rail **92** is a recessed surface **138** that provides additional clearance for the retractor blade **14** to slide onto the mounting rail **92**.

(69) With reference to FIGS. 4 and 6, the resilient arms **134, 136** engage the slot **220** of the blade **14**. The resilient arms **134, 136** have tapered surfaces **144, 146** that frictionally engage slot surfaces **223, 225** of the blade **14** and resist removal of the blade **14** from the mounting rail **92**. When the blade **14** is connected to the mounting rail **92**, the resilient arms **134, 136** are resiliently deflected toward each other in directions **140, 142** such that the resilient arms **134, 136** bias the tapered surfaces **144, 146** into frictional engagement with the slot surfaces **223, 225**.

(70) The guide dilator **10** and the retractor blades **12, 14, 16, 18** preferably have cooperating stop portions which limit sliding of the retractor blades beyond a position where the retractor blades **12, 14, 16, 18** are fully engaged with the mounting rails **20, 92, 94, 96**. For example, with reference to FIG. 4, the mounting rails **20, 94** have lower end walls **170, 172** which abut stop surfaces on the blades **12, 16**, such as the stop surface **174** of the blade **12** (see FIG. 5). The lower end walls **170,**

172 have curvatures that are complimentary to the stop surfaces of the blades **12, 16** which permits the surfaces to firmly engage the end walls **170, 172**. The mounting rails **92, 96** have similar lower end walls **178, 180** for engaging stop surfaces on the retractor blades **80, 82**, such as a stop surface **176** of the blade **14** (see FIG. 6).

(71) As shown in FIGS. 5 and 5A, a cannulated blade, such as the blade **12**, has a slot **190** for receiving the flange **112** of the mounting rail **20**. The slot **190** comprises an open end **192** and a closed end **194**, the mounting rail **20** being inserted into the open end **192** of the slot **190** and slid along the blade **12** until reaching the closed end **194** of the slot **190**. The blade **12** has inwardly extending walls **196, 198** which slide beneath the flange **112** and interlock therewith to restrict movement of the blade **12** radially outward from the guide dilator **10**. In one form, axially extending surfaces of the flange **112** and the walls **196, 198** extend from a midline **199** of the blade **12** toward the distal end **36** of the blade **12** a distance that is at least a quarter of a blade length **210** (see FIG. 5) when the blade **12** is connected to the guide dilator **10**. The walls **196, 198** have serrations **200, 202** for engaging ratchet arms **203, 205** of a docking anchor **201** (see FIG. 31) which may be inserted into the open end **192** of the slot **190** and slid down toward the closed end **194** with the ratchet arms **203, 205** resiliently biased into engagement with the serrations **200, 202** to releasably fix the depth of the docking anchor **201** within the blade **12**.

(72) Further, even if the retractor blades **12, 14, 16, 18** have varying straightness along their lengths, sliding the blades **12, 14, 16, 18** fully onto the mounting rails **20, 92, 94, 96** (see FIG. 1) holds distal ends **34, 35, 37, 39** of the blades **12, 14, 16, 18** to the guide dilator **10**. The interlocking engagement between the flanges of the mounting rails, such as the flange **112** of the mounting rail **20**, and the inwardly extending walls of the associated blade, such as walls **196, 198** of the blade **12**, ensures that the walls **196, 198** are held below the flange **112** when the stop surface **174** is slid into engagement with the lower end wall **170** of the mounting rail **20**. With the walls **196, 198** held firmly adjacent the lower base **110** of the mounting rail **20**, the blade distal end **34** extends from the lower end wall **170** of the mounting rail **20** along the leading end portion **42** of the guide dilator **10**. In this manner, the guide dilator **10** compensates for variation in the straightness of the retractor blades **12, 14, 16, 18** by ensuring the retractor blade distal ends **34, 35, 37, 39** are evenly spaced about the guide dilator **10** and spaced from each other.

(73) The blade length **210** is chosen for a particular operation and typically is in the range of 80 mm-180 mm for lateral approach surgery. As shown in FIG. 5, the blade **12** also has a tip distance **216** extending between the stop surface **174** and the distal end **34**. When the blade **12** is slid onto the mounting rail **20** and the lower end wall **170** of the mounting rail **20** abuts the stop surface **174** of the blade **12**, the blade **12** generally extends the tip distance **216** along the leading end portion **42** of the guide dilator **10** beyond the stop surface **174** (see FIG. 3).

(74) The blade **14** is substantially similar to the blade **12** with the exception of the cylindrical portion **64** and the cannula **74** thereof. Like the blade **12**, the blade **80** has a slot **220** with an open end **222**, a closed end **224**, and inwardly extending walls **226, 228** with serrations **230**. The blade **14** has a length **232** and a tip distance **234** extending between the stop surface **176** and a distal end **236** of the blade **14**. The guide dilator **10** may be provided with sets of blades **12, 14, 16, 18** having different blade lengths **210** or **232**, although the tip distance **216** or **234** remains similar for the different blade lengths. In this manner, the amount of leading end portion **42** covered by the blades **12, 14, 16, 18** is fixed for different sets of blades having different lengths.

(75) A guide dilator **300** in accordance with another form of the present invention is illustrated in FIG. 7. The guide dilator **300** is substantially similar to the guide dilator **10** previously discussed, with the exception that the guide dilator **300** has mounting rails **302, 304, 306, 308** with different mechanisms for retaining the respective retractor blades than the guide dilator **10**. More specifically, the mounting rails **302, 304**, which are intersected by grooves **310, 312**, have flanges **320, 322** with elongate sections **324, 326** and resilient members **328, 330** disposed in the middle of the elongate sections **324, 326**. The resilient members **328, 330** are resiliently biased radially

outward into engagement with associated retractor blades to provide a frictional engagement therebetween in a manner similar to the resilient arms **30**, **32**. Likewise, the mounting rails **306**, **308** have flanges **332**, **334** with resilient members **336**, **338** for creating frictional engagement with associated retractor blades.

(76) As shown in FIG. 7A, the resilient member **328** of the mounting rail **302** is spaced from a lower portion **329** of the rail **302** by a gap **331** when a retractor blade is not connected to the mounting rail **302**. Sliding a retractor blade onto the rail **302**, such as blade **12**, deflects the resilient member **328** radially inward in direction **333** and decreases the size of the gap **331**, as shown in FIG. 7B. With the retractor blade **12** connected to the rail **302**, the resilient member **328** biases against a surface **335** thereof radially outward in direction **337** into frictional engagement with a surface **158** of the blade **12** (see FIG. 5). In one approach, the guide dilator **300** is made of stainless steel or aluminum and the member **328** may resiliently flex due to the elongate gap spacing **331** machined into the mounting rail **302**. The retractor blade **12** may be made of, for example, aluminum or polyether ether ketone (PEEK).

(77) A guide dilator **410** and retractor blades **412**, **414** in accordance with another form of the present invention are shown in FIGS. 8-10. The guide dilator **410** is similar to the guide dilator **10** and has mounting rails **416**, **418** (see FIG. 9) for slidably engaging the blades **412**, **414**. Specifically, the blades **412**, **414** slide onto the mounting rails **416**, **418** in direction **430**, as shown in FIG. 8. For example, blade **412** slides in direction **430** into engagement with mounting rail **416** and may slide in direction **430** until a stop surface **433** (see FIG. 10) engages a lower end wall **432** of the mounting rail **416**. The blades **412**, **414** may be easily and rapidly slid onto or off of the guide dilator **10** along a leading end portion **434** thereof.

(78) Turning to FIG. 9, the blades **412**, **414** form a generally circular profile when engaged with the guide dilator **410**. One difference between the guide dilator **410** and the guide dilator **10** is that the guide dilator **410** has only two mounting rails **416**, **418** to accommodate blades **412**, **414** which each extend around the guide dilator **410** approximately 180 degrees. Another difference is that the blades **412**, **414** are both cannulated with a cylindrical portion **440**, **442** and cannulas **444**, **446** sized to receive a fixation pin, nerve probe, or other instrument as will be discussed in greater detail below.

(79) Referring next to FIGS. 11-24, a retraction procedure using the guide dilator **410** and the blades **412**, **414** is illustrated. Initially, an incision is made adjacent a desired surgical site **510** of a spine **512**. A dilator **516** is inserted in the incision into position at a surgical site **510**, such as into contact with the intervertebral disc **620**. A guide wire **514** is inserted into the cannula of the dilator **516** in direction **518** until the guide wire pierces into the intervertebral disc **620**, securing the position of the dilator **516**. As shown in FIG. 12, a second dilator **520** is slid downward along the first dilator **516** in direction **522** to further dilate the tissues surrounding the surgical site **510**. A third dilator **524** may be slid over the second dilator **520** to further expand the tissues, as shown in FIG. 13. Once the dilators **516**, **520**, **524** have initially enlarged the incision, the assembled guide dilator **410** and the retractor blades **412**, **414** may be slid along the dilator **524** in direction **526** into the incision. As is apparent, the leading end portion **434** of the guide dilator **410** enlarges the tissues surrounding the incision as the leading end portion **434** is advanced toward the surgical site **510**. Further, distal ends **540**, **542** of the retractor blades **412**, **414** also dilate the tissues as the distal ends **540**, **542** are advanced into the incision. In some approaches, it may be desirable to insert the guide dilator **410** and connected blades **412**, **414** as shown in FIG. 14 to permit retraction of the blades **412**, **414** in directions generally perpendicular to the length of the spine **512**.

(80) Once the leading end portion **434** of the guide dilator **410** reaches the surgical site **510**, the surgeon may want to monitor for the presence of adjacent nerves before the retractor blades **412**, **414** are slid downward along the leading end portion **434** into position adjacent the surgical site **510** would further dilate the tissues adjacent the surgical site **510**. One approach to nerve monitoring involves advancing a nerve monitoring probe **550** (see FIG. 15) along one of the

grooves 552, 554 extending along the length of the guide dilator 410 and into one of the cannulas 444, 446 (see FIG. 9) of the blades 412, 414. The nerve monitoring probe 550 has a shaft 556 ending at a bulbous tip 558 sized to fit into cannulas 444, 446 of the blades 412, 414. In the illustrated approach, the nerve monitoring probe 550 was inserted downward in direction 564 into the cannula 444 until the bulbous tip 558 extends downward beyond the tip 540 of the retractor blade 412. As shown in FIG. 17, the guide dilator 410 and the connected blades 412, 414 and nerve monitoring probe 550 may be rotated in directions 570, 572 to maneuver the tip 558 of the nerve monitoring probe 550 in a path around the surgical site 510. In this manner, the surgeon can monitor for the presence of nerves around the entirety of the leading end portion 434 before continuing the surgical procedure.

(81) If the results of the nerve monitoring are satisfactory, the retractor blades 412, 414 are slid downwardly down the guide dilator 410 in direction 588 until distal ends 540, 542 are positioned adjacent the surgical site 510, as shown in FIG. 18. The retractor blades 412, 414 remain engaged with the mounting rails 416, 418 as the retractor blades 412, 414 slide along the leading end portion 434 of the guide dilator 410. At this point, the surgeon may want to secure the retractor blade 412 to a vertebra 590 (see FIG. 22) to stabilize the blade 412 relative to the patient during subsequent steps of the retraction procedure. In the illustrated approach, a fixation pin 580 having a head 582, a shank 584, and a tip 586 (see FIG. 19) is driven in direction 588 into the cannula 444 of the blade 412 after the nerve monitoring probe 550 has been removed from the cannula 444. With the blade 412 secured to the vertebra 590, as shown in FIG. 20, a retractor 600 is moved along the guide dilator 410 in direction 602 until operating mechanisms 604, 606 reach the blades 412, 414. As will be discussed in greater detail below, the operating mechanisms 604, 606 may be secured with the blades 412, 414 by engaging dovetail connections between the operating mechanisms 604, 606 and the blades 412, 414 and fixing the operating mechanisms 604, 606 to the blades 412, 414.

(82) As shown in FIG. 21, the guide dilator 410, the guide wire 514, and the dilators 516, 520, 524 may be withdrawn from the incision in direction 610. With the operating mechanisms 604, 606 of the retractor 600 engaged with the blades 412, 414, the blades 412, 414 are held against movement while the guide dilator 410 is withdrawn in direction 610 outward from the incision. The relative movement between the guide dilator 410 and the retractor blades 412, 414 draws the mounting rails 416, 418 of the guide dilator 410 out of respective slots on the blades 412, 414 and disengages the connections therebetween. In this manner, withdrawing the guide dilator 410 simultaneously disengages the blades 412, 414 from the guide dilator 410 and removes the guide dilator 410 from a working channel 640 between the retractor blades 412, 414, as shown in FIG. 22.

(83) With reference to FIG. 22, the retractor 600 and the blades 412, 414 are shown positioned above the intervertebral disc 620 between vertebrae 590, 591. The operating mechanisms 604, 606 include sliders 622, 624 having dovetail recesses 626, 628 that slide over and into engagement with dovetail projections 630, 632 of the retractor blades 412, 414. The operating mechanisms 604, 606 have fixation screws 634, 636 that releasably fix the blades 412, 414 to the sliders 622, 624. As shown in FIG. 22, the retractor blades 412, 414 define an initial size of the working channel 640 to the intervertebral disc 620. To retract tissues in directions transverse to the operating mechanisms 604, 606, the retractor 600 has operating mechanisms 642, 644 that receive another pair of retractor blades such as a narrow retractor blade 650, as shown in FIG. 23. The narrow retractor blade 650 has a dovetail projection 652 and a body 654 extending from the dovetail projection 652 and ending at a tip 656. With reference to FIG. 24, narrow retractor blades 650, 662 have dovetail projections 652, 666 that are slid downward and engaged with the dovetail recesses 668, 670 with bodies 654, 674 advancing into gaps between the blades 412, 414. Fixation screws 676, 678 are tightened to engage the blades 660, 662 with the operating mechanisms 642, 644. The operating mechanisms 606, 642, and 644 are then retracted in directions 690, 692, 694 to enlarge the working channel 640 to a desired size to accommodate tools for operating on the intervertebral disc 620.

(84) With reference to FIG. 25, the operating mechanism 604 has a resilient element 700 disposed

in the dovetail recess **626** which aids in releasably securing the blade **630** to the operating mechanism **604**. More specifically, the operating mechanism **604** includes a wedge lock **633** having a fixation screw **634** threadingly engaged with a nut **635**, as shown in FIG. 26. The wedge lock **633** is disposed in a recess **643** of a slider inner member **641** of the operating mechanism **604**, as shown in FIG. 27. Tightening the fixation screw **634** draws the nut **635** upward along the fixation screw **634** and engages wedge **637** with an inclined surface **639** of an inner member **641**. Drawing the wedge **637** upward along the inclined surface **639** shifts the wedge outward in direction **645** which applies pressure against the dovetail projection **630** of the retractor blade **412** received in the dovetail recess **626** and fixes the blade **412** to the slider inner member **641**. Loosening the fixation screw **634** shifts the nut **635** downward along the screw **634** and the resilient element **700** biases the wedge **637** downward and radially inward in direction **647** to return the wedge **637** to an initial, unlocked position shown in FIG. 26.

(85) As shown in FIG. 25, the operating mechanism **644** includes a slider **710** disposed in a slot **712** of the retractor frame **714**. The operating mechanism **644** further comprises a ratchet mechanism **716** that permits the slider **710** to retract in direction **720** but restricts return movement of the slider **710** until a button **724** of the ratchet mechanism **10** is pressed to disengage the ratchet mechanism **716** from the slider **710**. With reference to FIG. 28, the slider **710** and portions of the ratchet mechanism **716** are shown removed from the frame **714** of the retractor **600**. The ratchet mechanism **716** includes an integral tooth **717** of the button **724** and a spring (not shown) that biases the button **724** in direction **719** around shaft **721** and the tooth **717** into engagement with depressions **723** on a base **735** of the slider **710**. Pressing the button **724** overcomes the biasing force of the spring and disengages the tooth **717** from the depressions **723** on the base **735**.

(86) With reference to FIG. 29, the slider **730** is shown removed from the retractor frame **714**. The slider **730** comprises an inner member **732** pivotally connected with a base **734** at a pivot pin **736**. The slider **730** includes an elevation screw **740** engaged with threads **741** of the inner member **732**. The elevation screw **740** has a ball **747** at one end thereof engaged with a socket **748** of the base **734**. Rotating the elevation screw **740** in direction **750** elevates an end **742** of the inner member **732** in direction **744** and rotation of the elevation screw **740** in an opposite direction lowers the end **742** in direction **746**. The connection between the ball **747** and the socket **748** permits the elevation screw **740** to rotate and elevate/lower the inner member **732** while pivoting relative to the base **734** as the inner member **732** elevates/lowers.

(87) With reference to FIG. 30, a retractor **800** in accordance with another form of the present invention is shown. The retractor **800** is substantially similar to the retractor **600**, except for operating mechanisms **802** that have a fixed pivot pin **804** about which a slider **806** of the operating mechanism **802** pivots in direction **808**. By contrast, the pivot pin **736** of the slider **750** (see FIG. 29) retracts with retraction of the slider **730**. Another difference is that the slider **806** has a slider outer body **810** which slides along axis **812** relative to a slider inner body **814**. Like the retractor **600**, the retractor **800** has an elevation screw **816** with a ball **818** and socket **820** for accommodating pivoting of the slider **806**. Further, the operating mechanism **802** includes a ratchet mechanism **822** that permits retraction of the slider **806** but restricts movement of the slider **806** in direction **826** in a manner similar to the operating mechanism **644**. The ratchet mechanism **822** includes a button **824** that is pressed to disengage the ratchet mechanism **822** and permit the slider **806** to move in direction **826**.

(88) With reference to FIG. 31, the docking anchor **201** has a body **207** sized to fit within a slot of a retractor blade, such as a slot **190** of the blade **12** (see FIG. 5). The body **207** is advanced into the open end **192** of the slot **190** and slid therein until reaching the closed end **194** of the slot. At this position, a blade **209** of the docking anchor **201** extends beyond the distal end **34** of the blade **12** such that a tip **211** of the blade **209** can penetrate a surface of an intervertebral disc during surgery. Further, the ratchet arms **203**, **205** of the docking anchor **201** are resiliently biased apart to engage serrations **200**, **202** of the blade to secure the docking anchor **201** to the blade **12** at a desired

position.

(89) As shown in FIG. 32, a disposable light source **900** for use with the retractor **600** is shown. The disposable light source **900** has a flexible fiber optic cable **902** and an anodized aluminum tip **904** sized to fit within a slot of a retractor blade, such as the slot **190** of retractor blade **60** shown in FIG. 5. The tip **904** includes a light source **906** for illuminating a retracted incision, such as the working channel **640** shown in FIG. 22.

(90) With reference to FIG. 33, a retractor **1010** in accordance with another aspect of the present invention is illustrated. The retractor **1010** is substantially similar to the retractor **600** such that the differences therebetween will be highlighted. The retractor **1010** has a frame **1012** including a raised portion **1014** that accommodates placement of the retractor **1010** adjacent a patient's iliac crest. The raised portion **1014** includes a pair of arms **1016**, **1018** supporting an operating mechanism **1020** disposed above the plane of the other operating mechanisms **1022**, **1024**, **1026**. The operating mechanism **1020** has a dovetail recess **1030** and a wedge lock **1032** operable via a screw **1034** similar to operating mechanism **604** discussed above; however, the operating mechanism **1010** lacks a slider to retract the associated retractor blade.

(91) As shown in FIG. 34, the retractor **1010** may be used to retract tissues adjacent lumbar vertebra **1040**, **1042**. The raised portion **1014** of the retractor **1010** provides spacing **1044** between the operating mechanism **1020** and an iliac crest **1046** of the patient. In this manner, a surgeon can position the raised portion **1014** upon a patient's hip while keeping the retractor **1010** relatively flush with the skin of the patient. The operating mechanism **1020** utilizes a longer blade **1050** to compensate for the distance the operating mechanism **1020** is elevated above the other operating mechanisms. Although blade **1050** is longer than the remaining blades **1052**, **1054**, **1056**, the blades **1050**, **1052**, **1054**, **1056** may all be inserted using a single guide dilator, such as guide dilator **10**, in a manner substantially similar to the processes described above. The blade **1050** has a longer slot than the blades **1052**, **1054**, **1056** so that the distal ends of the blades **1050**, **1052**, **1054**, **1056** will be disposed evenly at the leading end portion **46** of the guide dilator **10**.

(92) A pin dilator **1100** may also be used with the guide dilators **10**, **300**, **410** and surgical techniques described above. The pin dilator **1100** includes a dilator **1102** and a guide wire **1104** fixed in the dilator **1102**. The pin dilator **1100** is an easy-to-handle tool that may be used in place of a conventional guidewire **514** and first dilator **516** (see FIG. 11).

(93) Turning to FIGS. 36-41, a retractor **1200** in accordance with the present invention is shown. The retractor **1200** is similar in a number of ways to retractors **600** and **1010** and includes sliders **1212** connected to retractor blades **1214** which retract radially outward to enlarge an incision. The retractor **1200** includes a ratchet mechanism **1220** comprising a button **1222** and an integral tooth **1224** similar to the button **724** and tooth **717** (see FIG. 28) for selectively engaging the associated slider **1212**, as shown in FIG. 36. The retractor **1200** further includes a slider lock mechanism **1230** comprising a lock latch **1232** that is pivotally connected to the button **1222**. The slider lock mechanism **1230** fixes the tooth **1224** with the slider **1212** and reduces the chance that the ratchet mechanism **1220** may be accidentally disengaged during operation, which would permit the slider **1212** and associated retractor blade **1214** to shift radially inward. More specifically, when the slider **1212** as been retracted, a surgeon may pivot the lock latch **1232** toward a frame **1240** of the retractor **1200** such that the latch **1232** abuts the frame **1240** and restricts the button **1222** and integral tooth **1224** thereof from moving out of engagement with the slider **1212**, as shown in FIG. 37.

(94) The retractor **1200** also permits a surgeon to move the sliders **1212** radially inward to a beyond center position where the associated retractor blade **1214** connected to the slider **1212** is beyond a center of a working channel defined by the retracted retractor blades **1214**, as shown in FIG. 38. The retractor **1200** has a beyond center locking mechanism **1250** that controls whether a radially outer end **1252** of the slider **1212** may pass radially inward beyond the retractor frame **1240**. The beyond center locking mechanism **1250** includes a lock button **1254** shiftably connected to a base

1256 of the slider **1212** by a pin **1258**, as shown in FIGS. **39** and **40**. The pin **1258** travels in a slot **1260** of the base **1256** and permits the lock button **1254** to shift up and down as shown by arrow **1262** in FIG. **39**.

(95) For example, while retracting an incision, stray tissues may extend into the working channel between blades **1214**. In prior approaches, a surgeon would need to insert a tool into the surgical field and tuck the stray tissues behind the nearby retractor blades **1214** to remove the stray tissues from the working channel. By contrast, a surgeon utilizing the retractor **1200** who encounters stray tissues in the working channel, may simply shift the button **1254** upward so that a lower end **1270** is in clearance with a radially outer wall **1272** of the retractor frame **1240**, as shown in FIGS. **40** and **41**. The surgeon then may move the slider **1212** and retractor blade **1214** radially inward to the beyond center position, as shown in FIG. **38**. The slider **1212** and retractor frame **1240** include confronting surfaces **1280**, **1282** that abut and limit movement of the slider **1212** radially inward to a predetermined position along the retractor frame **1240**.

(96) With reference to FIG. **42**, a retractor **1300** is provided that is similar in many respects to the retractors discussed above. The retractor **1300** has a frame **1302** extending about a central opening **1304** and operating mechanisms **1306** for connecting tissue engaging members, such as retractor blades, to the frame **1302**. The operating mechanisms **1306** each include a slider **1308** to which a blade is connected and a ratchet mechanism **1310** for selectively resisting movement of the slider **1308** in direction **1314** toward an extended position thereof.

(97) The operating mechanism **1306** further includes a slider linear drive **1320** for driving the slider **1308** in direction **1312** opposite direction **1314** toward a retracted position thereof. Each operating mechanism **1306** may include an associated slider linear drive **1320**, which permits a surgeon to retract all of the sliders **1308** and blades connected thereto apart using the slider linear drives **1320** to enlarge an incision. The slider linear drive **1320** provides for ease in controlling retraction of the slider **1308** by using a tool engaged with the slider linear drive **1320** versus other systems that require use of tools that engage the blades directly for spreading the blades apart. These prior tools may be undesirable in some procedures because they are inserted between the blades and used to spread the blades apart, which involves positioning a portion of the tool in the working channel created by the retractor. It will be appreciated that although the slider linear drive **1320** provides advantages over tools that engage the blades directly for spreading the blades apart, the retractor **1300** may also be used with these prior tools in addition to or in place of the linear drive mechanism **1320** for retracting the blades.

(98) With reference to FIG. **43**, the slider linear drive **1320** includes a pinion **1330** with teeth **1332** that engage teeth **1334** of a rack **1336** of the slider **1308**. In one form, the slider **1308** has an inner member **1340** and an outer member **1342**. The inner member **1340** is pivotally connected to the outer member **1342** via a pivot member such as a pivot pin **1600**. The outer member **1342** includes the rack **1336** such that operation of the slider linear drive **1320** drives the outer member **1342** and inner member **1340** connected thereto in direction **1312** toward the retracted position thereof. The slider linear drive **1320** further includes a drive cap **1346** connected to the pinion **1330** at the upper end thereof as described hereinafter. The drive cap **1346** has a tool engaging portion, such as a socket **1348** configured to receive a hexalobular driver. In use, a surgeon may insert the hexalobular driver into the socket **1348** and turn the drive cap **1346** in a drive direction **1350**. This causes rotation of the pinion **1330** in drive direction **1350** and, by way of the engagement of the pinion teeth **1332** and the rack teeth **1334**, generates sliding movement of the slider **1308** in direction **1312** relative to the frame **1302** such that the slider **1308** slides toward the retracted position thereof.

(99) With reference to FIGS. **43** and **44**, the slider linear drive **1320** includes an anti-reverse device **1360** that limits rotation of the pinion **1330** to rotation in drive direction **1350**. By limiting rotation of the pinion **1330** to rotation in the drive direction **1350**, the retractor **1300** limits the likelihood of a surgeon using the hexalobular driver and the drive cap **1346** to unintentionally move the slider

1308 in direction **1314** and collapsing the working channel formed by the retracted blades.

(100) With reference to FIG. **44**, the drive cap **1346** and pinion **1330** are received in a bore hole **1370** of the frame **1302**. Both the drive cap **1346** and the pinion **1330** can rotate in drive direction **1350** within the bore hole **1370**. The drive cap **1346** has a groove **1384** extending thereabout that receives a split ring **1386**. The split ring **1386** extends into an annular recess **1388** extending about the bore hole **1370**. The split ring **1386** thereby retains the drive cap **1346** at a predetermined vertical position at an upper end of the bore hole **1370** while permitting the drive cap **1346** to rotate within the bore hole **1370**.

(101) In addition to being rotatable within the bore hole **1370**, the pinion **1330** can shift vertically in direction **1372** and disengage from the drive cap **1346** in response to a surgeon turning the drive cap **1346** in a reverse direction **1351** (see FIG. **43**). This disengagement inhibits the pinion **1330** from driving the slider **1308** in direction **1314** toward the extended position thereof even though the surgeon is rotating the drive cap **1346** in the reverse direction **1351**. More specifically, the anti-reverse device **1360** includes a washer **1374** abutting a lower end **1376** of the pinion **1330** and a washer **1378** abutting a seat **1380** of the frame **1302**, as shown in FIG. **44**. There is a biasing member, such as a spring **1382**, positioned between the washers **1374**, **1378**. The spring **1382** biases the washer **1374** upward and, in turn, biases the pinion **1330** upward in direction **1373** against the drive cap **1346**. The spring **1382** may be partially compressed when the drive cap **1346**, pinion **1330**, spring **1382**, and washers **1374**, **1378** are assembled within the bore hole **1370** which causes the spring **1382** to press the pinion **1330** against the underside of the drive cap **1346**.

(102) If the drive cap **1346** is rotated in reverse direction **1351**, the pinion **1330** is shifted downwardly in direction **1372** (see FIG. **44**) and disengages from the drive cap **1346** due to camming engagement between surfaces of the drive cap **1346** and pinion **1330**, as discussed in greater detail below. When the pinion **1330** shifts downwardly in direction **1372**, the pinion **1330** urges the washer **1374** toward the washer **1378** and further compresses the spring **1382**. The spring **1382** resiliently urges the pinion **1330** upwardly in direction **1373** in response to this compression.

(103) Because the surgeon's rotation of the drive cap **1346** in the reverse direction **1351** does not produce any movement of the slider **1308**, the surgeon should readily appreciate that the drive cap **1346** is being rotated in an incorrect direction and can then rotate the drive cap **1346** in the drive direction **1350**. As the surgeon rotates the drive cap **1346** in drive direction **1350**, the spring **1382** urges the pinion **1330** against the drive cap **1346**. The rotating drive cap **1346** eventually reaches the correct orientation relative to the pinion **1330** where the surfaces of the drive cap **1346** and pinion **1330** can mate and fully engage as discussed in greater detail below. The spring **1382** thereby helps return the drive cap **1346** and the pinion **1330** into mating engagement as the drive cap **1346** is rotated in the drive direction **1350** after the drive cap **1346** was rotated in the reverse direction **1351**.

(104) With reference to FIGS. **45** and **46**, the anti-reverse device **1360** includes cooperating protrusions of **1400**, **1402** of the pinion **1330** and protrusions **1404**, **1406** of the drive cap **1346**. When the drive cap **1346** is turned in drive direction **1350**, the protrusions **1400**, **1404** and **1402**, **1406** mate and fully engage which causes turning of the pinion **1330** in drive direction **1350**. When the drive cap **1346** is turned in the reverse direction **1351**, the protrusions **1404**, **1406** disengage from the protrusions **1400**, **1402** and ride along the protrusions **1400**, **1402**. Even though the drive cap **1346** may be turned in the reverse direction **1351**, this turning does not translate into turning of the pinion **1330** in the reverse direction **1351** due to the disengagement of the protrusions **1404**, **1406** from the protrusions **1400**, **1402**.

(105) More specifically and with reference to FIG. **45**, the protrusions **1400**, **1402** of the pinion **1330** include generally vertically extending drive surfaces **1410**, **1412** and cam surfaces **1418**, **1420** that ramp up to upper ends of the drive surfaces **1410**, **1412**. The pinion **1330** includes lands **1414**, **1416** extending circumferentially away from the drive surfaces **1410** and which lead into the cam surfaces **1418**, **1420**. With reference to FIG. **46**, the protrusions **1404**, **1406** of the drive cap **1346**

likewise have generally vertically extending drive surfaces **1422**, **1424** and cam surfaces **1430**, **1434** that ramp downwardly to lower ends of the drive surfaces **1422**, **1424**. The drive cap **1346** has lands **1426**, **1428** extending circumferentially away from the drive surfaces **1422**, **1424** and which lead into the cam surfaces **1430**, **1434**. It is noted that FIG. **46** shows the drive cap **1346** in an upside-down orientation such that the cam surfaces **1430**, **1434** are shown ramping upward toward the drive surfaces **1422**, **1424**, but the cam surfaces **1430**, **1434** would be oriented to ramp downward toward the drive surfaces **1422**, **1424** when the drive cap **1346** and pinion **1330** are assembled.

(106) When the drive cap **1346** and the pinion **1330** are positioned in the frame bore hole **1370**, the protrusions **1404**, **1406** of the drive cap **1346** are positioned adjacent the lands **1414**, **1416** of the pinion **1330**. When the surgeon inserts the hexalobular driver into the socket **1348** of the drive cap **1346** and turns the drive cap **1346** in drive direction **1350**, this causes the drive surface **1422** of the drive cap **1346** to abut the drive surface **1410** of the pinion **1330** and causes the drive surface **1424** of the drive cap **1346** to abut the drive surface **1412** of the pinion **1330**. Because of the abutting drive surfaces **1410**, **1422** and **1412**, **1424**, turning of the drive cap **1346** in the drive direction **1350** produces corresponding turning of the pinion **1330** in drive direction **1350** and causes retraction of the slider in direction **1312**.

(107) Conversely, if the surgeon rotates the hexalobular driver engaged with the drive cap **1346** in the reverse direction **1351**, the cam surface **1430** of the drive cap **1346** cammingly engages the cam surface **1418** of the pinion **1330** and the cam surface **1434** of the drive cap **1346** cammingly engages the cam surface **1420** of the pinion **1330**. This camming engagement between the surfaces **1430**, **1418** and **1434**, **1420** shifts the pinion **1330** downward in direction **1372** (see FIG. **44**) within the bore hole **1370** and compresses the spring **1382**. The downward movement of the pinion **1330** permits the projections **1404**, **1406** of the drive cap **1346** to snap past the projections **1400**, **1404** of the pinion **1330** as the drive cap **1346** is turned in the reverse direction **1351** thereby inhibiting the drive cap **1346** from causing similar turning of the pinion **1330** in reverse direction **1351**. In effect, turning the drive cap **1346** in reverse direction **1351** moves the drive surfaces **1422**, **1424** of the drive cap **1346** away from the drive surfaces **1410**, **1412** of the pinion **1330** whereas turning the drive cap **1346** in drive direction **1350** brings the drive surfaces **1422**, **1424** into engagement with the drive surfaces **1410**, **1412**. Because the drive surfaces **1410**, **1422** and **1412**, **1424** do not engage when the drive cap **1346** is turned in reverse direction **1351**, the drive cap **1346** is unable to cause rotation of the pinion **1330** in the reverse direction **1351**.

(108) With reference to FIGS. **47-50**, the ratchet mechanism **1310** includes an actuator, such as the actuator button **1450**, which is similar to the actuator buttons **724**, **824**, **1222** discussed above. The button **1450** includes a projection, such as an integral projection tooth **1452**, which is similar to the projection teeth **717**, **1224** discussed above. The tooth **1452** engages recesses, such as depressions **1454**, between adjacent teeth **1454A** on a side of the slider outer member **1342** opposite the rack **1336**. The depressions **1454** are similar to the depressions **723** discussed above and both recesses receive a cooperating tooth **1452**, **717**. The button **1450** has a through opening **1456** that receives a shaft **1458** rotatably mounted at its ends to the frame **1302** for pivotally mounting the button **1450** to the frame **1302**. The ratchet mechanism **1310** further includes a slider lock mechanism **1460** that reduces the chance of the ratchet mechanism **1310** being accidentally disengaged during operation which would permit the slider **1308** and associated retractor blade to shift radially inward. The slider lock mechanism **1460** includes a lock latch **1462** pivotally connected to the button **1450** by a pin **1464**.

(109) With reference to FIGS. **49**, **51**, and **53** the lock latch **1462** may be pivoted between an unlocked position (see FIG. **49**), a ratchet position (see FIG. **51**), and a locked position (see FIG. **53**) for limiting operation of the button **1450** as discussed in greater detail below. The lock latch **1462** has an orientation indicia **1470** that may be used to provide visual or tactile feedback to the surgeon regarding the position of the lock latch **1462**. Further, the frame **1302** may have

corresponding indicia, such as an unlocked indicia **1472** (see FIG. 49), a ratchet indicia **1474** (see FIG. 51), and a locked indicia **1476** (see FIG. 51) which indicate that the lock latch **1462** is in a desired position once the orientation indicia **1470** is aligned with the respective indicia **1472**, **1474**, **1476**.

(110) Returning to FIG. 47, the button **1450** and the lock latch **1462** include interdigitated loops **1480**, **1482** extending about the pin **1464**. The loops **1480**, **1482** and pin **1464** form a hinge that permits pivoting of the lock latch **1462** relative to the button **1450**. The lock latch **1462** includes a handle portion **1484** that may include a scalloped outer surface to aid gripping by the surgeon and a tool receiving portion **1486**. As an example, the tool receiving portion **1486** may have a socket for receiving a hexalobular driver so that the surgeon may pivot the lock latch **1462** using the hexalobular driver. The hexalobular driver may be used when tissue pressure against the blades requires more persuasion to unlock the slider **1308** than the surgeon can apply with her finger tip(s).

(111) The lock latch **1462** further includes a projection, such as a tab portion **1490**, extending from the handle portion **1484** that includes an end **1492**. The lock latch surface **1494** abuts against surfaces of the frame **1302** when the lock latch **1462** is in the unlocked and locked positions, as discussed in greater detail below. With reference to FIGS. 47 and 48, the button **1450** includes a blind bore **1500** for receiving one end of a spring **1502**. The other end of the spring **1502** is received in a recess **1504** of the frame **1302**. With reference to FIG. 48, the spring biases the button **1450** in direction **1506** about the shaft **1458** and urges the tooth **1452** of the button **1450** into engagement with the depressions **1454** of the slider outer member **1342**. As with the ratchet mechanisms described above, the movement of the slider in direction **1312** toward the retracted position thereof causes the tooth **1452** to ratchet over the teeth **1454A** of the slider **1308** and permits the slider **1308** to retract. Conversely, the spring **1458** urges the tooth **1452** into engagement with the depressions **1454** and the contact between the tooth **1452** of the button **1450** and the tooth **1454A** of the slider **1308** adjacent the depression **1454** resists movement of the slider **1308** in direction **1314** toward the extended position thereof.

(112) Turning to FIG. 47A, the button **1450** includes a detent mechanism **1471** for holding the lock latch **1462** in the unlocked, ratchet, or locked positions thereof and resisting unintentional pivoting of the lock latch **1462** out of the selected position. The detent mechanism includes a ball **1473** and a spring **1475** received in a blind bore of the button **1450**. The spring **1475** resiliently urges the ball **1473** partially outward from the blind bore and into engagement with an indexing surface **1469** of the lock latch **1462**. The indexing surface **1469** includes grooves **1477**, **1479**, and **1481** that correspond, respectively, to the locked, ratchet, and unlocked positions of the lock latch **1462**. For example, once the lock latch **1462** has been pivoted to the ratchet position thereof, the ball **1473** snaps into the groove **1479**. To turn the lock latch **1462** to the locked position thereof, the surgeon applies sufficient force to overcome the biasing force from the spring **1475** and cam the ball **1473** out of the groove **1479**. Once the lock latch **1462** reaches the locked position, the spring **1475** urges the ball **1473** into the groove **1477**.

(113) With reference to FIGS. 48 and 49, the operation of the lock latch **1462** will be discussed in greater detail. The frame **1302** includes a slot **1520** that receives the slider **1308**. The slot **1520** is formed by a pair of spaced walls **1522**, **1524** on opposite sides of the slider **1308** that guide the slider **1308** as the slider **1308** shifts in directions **1312**, **1314**. The wall **1522** has an opening **1525** in communication with the bore hole **1370** through which the teeth **1332** of the pinion **1330** extend and engage the teeth **1334** of the slider rack **1336**. The other wall **1524** includes a through opening **1526** and the tooth **1452** extends therethrough to engage the depressions **1454** of the slider **1308**. The frame **1302** includes a cavity **1530** near the wall **1524** with a flat floor surface **1532** and walls **1534** upstanding from the floor surface **1532** that are arranged in a generally U-shaped configuration. The lower end of the lock latch tab portion **1490** is received in the cavity **1530** and pivots therein with pivoting of the lock latch **1462** between the unlocked, ratchet, and locked

positions thereof.

(114) With reference to FIG. 49, the lock latch **1462** has been pivoted to the unlocked position with the orientation indicia **1470** of the lock latch **1462** aligned with unlocked indicia **1472** of the frame **1302**. Pivoting the lock latch **1462** to the unlocked configuration brings the end **1492** of the lock latch **1462** into abutting contact with a surface **1540** of the walls **1534** of the cavity **1530** and pivots the button **1450** in direction **1560**. This causes the button **1450** to compress the spring **1502** and hold the tooth **1452** out of engagement with the depressions **1454** of the slider **1308**. Further, the lock latch **1462** resists pivoting of the button **1450** in direction **1506** about the shaft **1458** and continues to cause the button **1450** to hold the tooth **1452** out of engagement with the depressions **1454** until the lock latch **1462** is pivoted away from the unlocked position. Thus, with the lock latch **1462** in the unlocked position, the surgeon can shift the slider **1308** in direction **1312** or **1314** as desired without the tooth **1452** resisting movement of the slider **1308**.

(115) With reference to FIGS. 50 and 51, the lock latch **1462** has been pivoted to the ratchet position thereof such that the orientation indicia **1470** of the lock latch **1462** is aligned with the ratchet indicia **1474** of the frame **1302**. Because the end **1492** of the lock latch **1462** no longer abuts the cavity surface **1540**, the spring **1502** can pivot the button **1450** in direction **1506** about the shaft **1458** and cause the tooth **1452** to engage the depressions **1454** of the slider outer member **1342**. The engaged tooth **1452** and depressions **1454** are configured to permit the slider **1308** to slide in direction **1312**. When the slider **1308** slides in direction **1312**, the tooth **1452** received in one depression **1454** cams over and past the adjacent tooth **1454A**, ratchets into the next depression **1454**, cams over and past the next tooth **1454A**, ratchets into the next depression **1454**, cams over the next tooth **1454A**, etc. However, the tooth **1452** can abut flat surfaces **1454B** of the teeth **1454A** and resist movement of the slider **1308** in direction **1314**.

(116) To move the slider **1308** in direction **1314** when the lock latch **1462** is in the ratchet position, the surgeon presses on a side **1511** (see FIG. 51) of the button **1450** and pivots the button **1450** in direction **1560**. This compresses the spring **1502** and withdraws the tooth **1452** out of engagement with the depressions **1454**. With the tooth **1452** disengaged from the depressions **1454**, the slider **1308** may be slid in direction **1314** toward the extended position thereof.

(117) With the lock latch **1462** in the ratchet position of FIGS. 50 and 51, the lock latch **1462** can move back-and-forth within the cavity **1530** in directions **1560**, **1506** with the button **1450**. Specifically, the lock latch **1462** pivots in direction **1560** as the button **1450** pivots in direction **1560** about the shaft **1458** due to the tooth **1452** being cammed out of the way by the inclined surface of one of the teeth **1454A** as the slider **1308** shifts in direction **1312**. For example, a surgeon may insert a hexalobular driver into the socket **1348** of the drive cap **1346** and turn the drive cap **1346** and pinion **1330** in drive direction **1350**. This causes the slider **1308** to shift in direction **1312** toward the retracted position thereof and the inclined surface of the one tooth **1454A** cams the tooth **1452** outward and out of engagement with the depression **1454**. This camming action causes the button **1450** to pivot in direction **1560** and the lock latch **1462** likewise pivots with the button **1450** in direction **1560** within the cavity **1530**.

(118) Once the tooth **1452** has been cammed out of the way of the one tooth **1454A**, the button **1450** can pivot back in direction **1506** and advance the tooth **1452** into engagement with the next depression **1454** as the slider **1308** slides in direction **1312**. The lock latch **1462** likewise pivots with the button **1450** in direction **1506** within the cavity **1530**. Thus, the lock latch **1462** moves back-and-forth in directions **1560**, **1506** as the slider **1308** is shifted in direction **1312** and the tooth **1452** ratchets in and out of engagement with the depressions **1454**. The cavity **1530** provides clearance for this back-and-forth movement of the lock latch **1462**.

(119) Upon the slider **1308** reaching the desired retracted position, the surgeon may pivot the lock latch **1462** to the locked orientation shown in FIGS. 52 and 53. With the lock latch **1462** pivoted to the locked orientation, the orientation indicia **1470** of the lock latch **1462** is aligned with the lock indicia **1476** of the frame **1302** and the end **1492** of the lock latch tab portion **1490** abuts a surface

1570 of the wall 1524. Due to the lock latch end 1492 abutting the wall surface 1570, the tab portion 1490 rigidly resists pivoting of the button 1450 in direction 1560 which would disengage the tooth 1452 from the depressions 1454. Thus, with the lock latch 1462 in the locked position, the lock latch 1462 maintains the tooth 1452 engaged with one of the depressions 1454 of the slider outer member 1342. The lock latch 1462 thereby resists contact against the button 1450 causing the button 1450 to withdraw the tooth 1452 from the depression 1454 and permitting the slider 1308 to shift in direction 1314 toward its extended position. Thus, even if the surgeon were to press against the side 1511 and attempt to pivot the button 1450 in direction 1560, the lock latch 1462 would rigidly resist the pivoting of the button 1450 and would thereby keep the tooth 1452 engaged with one of the depressions 1454.

(120) When the surgeon wants to shift the slider 1308 in direction 1314, the surgeon simply pivots the lock latch 1462 back to the ratchet orientation (see FIG. 51) and presses against the side 1511 of the button 1450 to disengage the tooth 1452 from the depression 1454. Alternatively, the surgeon may pivot the lock latch 1462 to the unlocked orientation (see FIG. 49) which automatically pivots the button 1450 in direction 1560 and withdraws the tooth 1452 from the depression 1454.

(121) With reference to FIGS. 54 and 55, the inner member 1340 is pivotally connected to the outer member 1342 at the pivot pin 1600. The slider 1308 includes an elevation screw 1602 having a threaded shank 1604 engaged with a threaded recess 1606 of the inner member 1340. The elevation screw 1602 has a distal ball 1608 that is received in a socket 1610 of the outer member 1342. The socket 1610 holds the ball 1608 and permits pivoting of the ball 1608 relative to the outer member 1342. As the elevation screw 1602 is turned, the engagement between the elevation screw 1602 and the inner member 1340 pivots the inner member 1340 about the pivot pin 1600 to an inclined orientation as shown in FIG. 55. The pocket 1610 may be elongated along the slider outer member 1342 and include a pocket surface 1620 engaged with the ball 1608 that permits the ball 1608 to shift in directions 1622 as the inner member 1340 pivots relative to the outer member 1342.

(122) The slider 1308 also includes a wedge lock 1630 for releasably securing a blade within a dovetail recess 1632 of the inner member 1340. The wedge lock 1630 includes a wedge member 1634 pivotally mounted on a pin 1636 within a recess 1640 of the inner member 1340. The wedge lock 1630 further includes a fixation screw 1642 having a head 1644 for receiving an adjustment tool and a shank 1646 threadingly engaged with a nut 1648. The nut 1648 is connected to the wedge 1634 and configured to transfer vertical movement of the nut 1648 along the shank 1646 into pivoting of the wedge 1634. For example, the nut 1648 may have a pair of pins extending outward therefrom that are received in elongated slots of the wedge member 1634. The slots of the wedge member 1634 are configured to cammingly engage the pins of the nut 1648 and transfer linear movement of the nut 1648 into pivoting of the wedge member 1634.

(123) To secure a blade within the dovetail recess 1632 of the inner member 1340, a surgeon may rotate the fixation screw 1642 in a clockwise direction (when viewed from above) and cause the fixation screw 1642 to draw the nut 1648 upward in direction 1650 and pivots the wedge member 1634 in direction 1652. Pivoting the wedge member 1634 in direction 1652 causes an outer surface 1654 of the wedge member 1634 to press against the blade and lock the blade within the dovetail recess 1632. Conversely, the surgeon may rotate the fixation screw 1642 in a counter-clockwise direction (when viewed from above) and cause the nut 1648 to travel downwardly in direction 1660 and pivot the wedge member 1634 in direction 1662 to disengage the wedge member outer surface 1654 from the blade.

(124) The slide 1308 also includes a beyond center locking mechanism 1670 including a lock button 1672 having a lower end 1674 for interfering with the retractor frame 1302 and limiting movement of the slider 1308 in direction 1314 to a predetermined extended position. The beyond center locking mechanism 1670 further includes a pin 1676 received in a recess 1678 of the slider outer member 1342 to connect the lock button 1672 to the outer member 1342. The recess 1678 is elongated and permits the surgeon to shift the lock button 1672 in direction 1680 to raise the button

lower end **1674** out of interference with the frame **1302** and permit the slider **1308** to shift further in direction **1314** toward a beyond center position thereof.

(125) The components of the retractor **1300** may be made from a variety of materials. As but some examples in this regard, the frame **1302** may be made from a metallic or plastic material, such as aluminum or carbon fiber. The material of the frame **1302** may be selected such that the frame **1302** is radiolucent. The slider **1308** may be made from a metallic material, such as stainless steel, aluminum, or titanium. The button tooth **1452** may be made from a metallic material such as stainless steel. The lock latch **1462** may be made from a plastic material, such as an instrument grade polyaryletherketone.

(126) With reference to FIG. **56**, a retractor **1800** is provided that is similar in many respects to the retractors **600**, **1010**, **1210**, **1300** discussed above. The retractor **1800** includes a frame **1802** extending about a central opening **1804** and having one or more operating mechanisms **1806** that connect tissue engaging members such as retractor blades **1214** to the frame **1802**. Each operating mechanism **1806** includes a slider **1810** and a slider linear drive **1812** for shifting the slider **1810** between extended and retracted positions relative to the frame **1802**. The slider linear drive **1812** includes a tool-receiving member, such as a worm screw **1814**, disposed in a bore hole **1816** of the frame **1802**. Regarding FIG. **57**, turning the worm screw **1814** in a clockwise direction **1820** causes the slider **1810** to retract in direction **1822**. Conversely, turning the worm screw **1814** in a counterclockwise direction **1824** causes the slider **1810** to shift in direction **1826**.

(127) With reference to FIG. **57**, the worm screw **1814** is engaged with a worm gear **1834** of a worm gear assembly **1836**. The worm gear assembly **1836** includes a rotatable drive member, such as a pinion gear **1838**, that is selectively coupled to the worm gear **1834**. The worm gear **1834** has a helical tooth **1874** (see FIG. **58**) that engages teeth **1870** of the worm gear **1834** (see FIG. **59**). The worm screw **1814** and the worm gear **1834** are configured to be self-locking to inhibit unintended rotation of the worm gear **1834** and pinion gear **1838** coupled thereto which locks the slider **1810** in position relative to the frame **1802** when the surgeon is not turning the worm screw **1814**.

(128) Further, the engagement between the helical tooth **1874** and teeth **1870** permits positioning of the slider **1810** that is not limited by increments defined by a ratchet mechanism. The continuous helical tooth **1874** of the worm screw **1814** is always engaged with at least one of the teeth **1870** of the worm gear **1834** and operates as an inclined plane to drive the at least one tooth **1870** the distance imparted by turning of the worm screw **1814**. In other words, the engagement between the helical tooth **1874** and teeth **1870** permits infinitely adjustable positioning of the slider **1810** and tissue engaging member. A user can thereby turn the worm screw **1814** to cause as large or as small of a change in position of the slider **1810** as desired. Because the slider **1810** is not limited to shifting in increments dictated by a ratchet mechanism, the retractor **1800** provides enhanced flexibility in positioning the sliders **1810** with respect to the anatomy of a patient.

(129) With reference to FIG. **57**, each operating mechanism **1806** includes a clutch **1830** having a release mode wherein the clutch **1830** decouples the worm gear **1834** from the pinion gear **1838**. The clutch **1830** is similar to the clutch discussed above that includes the cam surface **1418** of the pinion **1330** and the cam surface **1434** of the drive cap **1346**. Turning the drive cap **1346** in reverse direction **1351** reconfigures the clutch to a release mode, causes the surfaces **1418**, **1434** to cammingly engage, and shifts the pinion **1330** away from the drive cap **1346**. The drive cap **1346** can thereby turn in reverse direction **1351** relative to the pinion **1330**. Likewise, when the clutch **1830** is in the release mode, the pinion gear **1838** can turn relative to the worm gear **1834** which permits the slider **1810** to be shifted by a user in direction **1822** or direction **1826** without resistance from the meshed worm screw **1814** and worm gear **1834**. The clutch **1830** also includes a drive mode wherein the clutch **1830** couples the worm gear **1834** and the pinion gear **1838**. With the clutch **1830** in the drive mode, turning of the worm screw **1814** in directions **1820**, **1824** causes turning of the worm gear **1834** and pinion **1838** coupled therewith and associated shifting of the slider **1810** in directions **1822**, **1826**.

(130) Regarding FIG. 57, the slider **1810** has a base, such as outer member **1842**, and an arm, such as inner member **1844**, pivotally coupled to the outer member **1842** by a pivot pin **1846**. The outer member **1842** includes a rack **1850** having teeth **1852** that are meshed with teeth **1854** of the pinion gear **1838**. Thus, when the clutch **1830** is in the drive mode, rotating the worm screw **1818** in direction **1820** causes rotation of the pinion gear **1838** in direction **1840** which shifts the slider outer member **1842** in direction **1822**. Conversely, turning the worm screw **1818** in direction **1824** causes rotation of the pinion gear **1838** in direction **1860** which shifts the slider outer member **1842** in direction **1826**. Thus, the worm screw **1814** is turned to shift the slider **1810** in either the retracting direction **1822** or the release direction **1826**.

(131) With reference to FIGS. 58 and 59, the teeth **1870** of the worm gear **1834** and the helical tooth **1874** of the worm screw **1814** are configured to be self-locking. In other words, the frictional engagement between one or more of the teeth **1870** and the helical tooth **1874** is sufficiently high to resist turning of the worm gear **1834** due to force imparted on a tissue engagement member supported by the slider **1810**. Thus, once the user has driven the worm screw **1814** to position the slider **1810** in the desired position, the slider **1810** is locked in position and cannot shift in either direction **1822**, **1826** until the surgeon decides to again drive the worm screw **1814** in direction **1820** or **1824** or reconfigures the clutch **1830** from the drive mode to the release mode.

(132) With reference to FIG. 59, the teeth **1870** have each have a thread profile with a generally involute gear shape and the helical tooth **1874** has a trapezoidal thread profile. The self-locking operation of the worm screw **1814** and the worm gear **1834** are provided at least in part by surfaces **1890**, **1892** of each of the teeth **1870** and surfaces **1894**, **1896** of the helical tooth **1874**. For example, once the slider **1810** has been retracted to a desired position, the retracted tissue may urge the slider **1810** in direction **1826** (see FIG. 57). This urges the tooth **1870** in direction **1860** (see FIG. 59) and presses the surface **1892** of the worm gear tooth **1870** against the upper surface **1894** of the helical tooth **1874**. The engagement between the surfaces **1892**, **1894** resists turning of the worm gear **1834** in direction **1860**. Because the worm gear **1834** does not turn in direction **1860**, and the clutch **1830** is in the drive mode, the pinion gear **1838** does not turn in direction **1860** and the slider **1810** does not shift in direction **1826** despite the loading from the retracted tissue.

(133) As another example, once the slider **1810** has been retracted to a desired position, if an instrument contacts the tissue engaging member and urges the slider **1810** in direction **1822**, the pinion **1838** will be urged in direction **1840** which would bring the surface **1890** (see FIG. 59) into engagement with the surface **1896** of the helical tooth **1874**. The engagement between the surfaces **1890**, **1896** inhibits turning of the worm gear **1834** in direction **1840**. Because the worm gear **1834** does not turn in direction **1840**, and the clutch **1830** is in drive mode, the pinion gear **1838** does not turn in direction **1840** and the slider **1810** does not shift in direction **1822** despite the contact with the instrument.

(134) Regarding FIG. 59, the worm screw **1814** includes a worm portion **1872** including the helical tooth **1874**, a head **2000**, and a shank **2002**. The head **2000** includes a tool-engaging portion, such as a socket **2004**, configured to receive a hexalobular driver to turn the worm screw **1814**. The worm screw **1814** is captured in the bore hole **1816** (see FIG. 56) of the frame **1802** by a c-ring positioned above a flange **2006** of the worm screw **1814**. The worm screw **1814** is rotatable about an axis **2008** and is subjected to loading parallel and transverse to the axis **2008** due to engagement with the worm gear **1834**. In one embodiment, thrust bearings **2010**, **2012** support the worm screw **1814** in the bore hole **1816** and permit the worm screw **1814** to turn with limited resistance from the frame **1802**.

(135) With reference to FIGS. 56 and 58, the clutch **1830** is operable to disengage the pinion gear **1838** from the worm gear **1834**. In one embodiment, the clutch **1830** is a component of the worm gear assembly **1836** and turns with turning of the worm screw **1814**. More specifically, the clutch **1830** includes a button **2020** that is pressed in direction **2022** to decouple the worm gear **1834** from the pinion gear **1838** and reconfigure the clutch **1830** from the drive mode to the release mode.

When the clutch **1830** is in the release mode, the pinion gear **1838** may turn relative to the worm gear **1834**. Because the pinion gear **1838** may turn relative to the worm gear **1834**, the slider **1810** may be shifted in directions **1826**, **1822** to a desired position without resistance from the intermeshed worm gear **1834** and the worm screw **1814**. The surgeon may shift the slider **1810** by hand in directions **1826**, **1822** or by utilizing an instrument when the clutch **1830** is in the release mode.

(136) With reference to FIG. **60**, the clutch **1830** includes a release shaft **2030** to which the button **2020** is mounted, such as via a threaded connection therebetween. The release shaft **2030** has a collar **2032**, a cylindrical portion **2034**, a neck portion **2036**, and a head portion **2038**. The clutch **1830** includes a spring **2040**, a spring seat washer **2042**, and a retaining ring **2044**. The head **2038** of the release shaft **2030** engages at least one coupling member, such as ball bearings **2048**. The ball bearings **2048** are received in openings **2050** of the pinion gear **1838** and may extend into pockets **2102** of the worm gear **1834**.

(137) The ball bearings **2048** are shiftable radially between a radially outward position wherein the ball bearings **2048** extend into the worm gear pockets **2102** and couple the pinion **1838** to the worm gear **1834** and a radially inward position where the ball bearings **2048** are spaced from the worm gear **2034** and decouple the pinion gear **1838** from the worm gear **1834**. To reconfigure the clutch **1830** from the drive mode to the release mode, the user presses the button **2020** in direction **2022** (see FIG. **61B**) which shifts the release shaft **2030** in direction **2022** to position the neck portion **2036** in alignment with the ball bearings **2048** thereby permitting the ball bearings **2048** to shift radially inward as discussed in greater detail below. A snap ring **2037** (see FIG. **60**) may be provided to capture the worm gear assembly **1836** within a bore hole **1839** (see FIG. **56**) of the frame **1802**.

(138) With reference to FIG. **61A**, the release shaft **2030** is shown in a drive position thereof. The worm gear **1834** includes a bore **2080** that receives the cylindrical portion **2034** of the release shaft **2030** and the spring **2040** extending thereabout. The worm gear **1834** includes a wall **2082** extending about the bore **2080** that includes a groove **2084** which receives the retaining ring **2044**. The spring **2040** is captured between the collar **2032** of the release shaft **2030** and the spring seat washer **2042**. Thus, pressing the button **2020** in direction **2022** shifts the release shaft **2030** in direction **2022** and brings the collar **2032** closer to the spring seat washer **2042** which compresses the spring **2040**. The button **2020** is pressed to shift the release shaft **2030** in direction **2022** until the release shaft **2030** reaches a release position thereof which radially aligns the neck portion **2036** of the release shaft **2030** with the ball bearings **2048**. The neck portion **2036** permits the ball bearings **2048** to shift radially inward which decouples the worm gear **1834** from the pinion gear **1838**. When the user releases the button **2020**, the spring **2040** urges the release shaft **2030** back in direction **2090** to the drive position thereof wherein the head portion **2038** of the release shaft **2030** is radially aligned with the ball bearings **2048**. The spring **2040** thereby automatically returns the clutch **1830** to the drive mode once the user releases the button **2020**.

(139) Regarding FIGS. **60** and **61A**, the worm gear **1834** includes a sleeve portion **2091** extending about an annular wall **2093** of the pinion **1834**. The annular wall **2093** has the through openings **2050** extending therethrough. The sleeve portion **2091** of the worm gear **1834** has pockets **2102** spaced thereabout. The pockets **2102** include openings **2103** sized to permit a portion of the ball bearings **2048** to extend into the pockets **2102**.

(140) With reference to FIGS. **61A** and **62A**, the release shaft **2030** is shown in the drive position and the head portion **2038** radially aligned with the ball bearings **2048**. The head portion **2038** of the release shaft **2030** has a diameter **2092** that is larger than a diameter **2094** of the neck portion **2036**. The larger diameter **2092** of the head portion **2038** keeps the ball bearings **2048** extending into the pockets **2102** of the worm gear **1834** and inhibits the ball bearings **2048** from shifting radially inward. When the smaller diameter **2094** of the neck portion **2036** is aligned with the ball bearings **2048**, the clearance between the neck portion **2036** and the ball bearings **2048** permits the

ball bearings **2048** to shift radially inward and out of the pockets **2102** of the worm gear **1834**. (141) With reference to FIGS. **61B** and **62B**, the release shaft **2030** is shown in the release position and the neck portion **2036** is radially aligned with the ball bearings **2048**. When the pinion **1838** is urged in one of the directions **1840**, **1860**, such as by a user manually shifting the slider **1810** in direction **1822** or direction **1826**, the pinion gear **1838** will turn and curved surfaces **2100** of the worm gear pockets **2102** cam the ball bearings **2048** radially inward in direction **2104**.

(142) Once the surgeon has moved the slider **1810** to the desired position and has released the button **2020**, the spring **2040** urges the release shaft **2030** in direction **2090** to bring the head portion **2038** of the release shaft **2030** into the release position. With reference to FIG. **61A**, the release shaft **2030** has a cam surface **2093** that is configured to engage and shift the ball bearings **2048** radially outward in direction **2112** as the spring **2040** returns the release shaft **2030** to the drive position thereof. Once the release shaft **2030** is in the drive position, the head portion **2038** keeps the ball bearings **2048** from shifting radially inward in direction **2104**.

(143) As shown in FIG. **62A**, the ball bearings **2048** extend in the pockets **2102** of the worm gear **1834** and the through openings **2050** of the pinion **1838**. The ball bearings **2048** are therefore subjected to shearing forces from the worm gear **1834** and the pinion gear **1838** as the ball bearings **2048** lock the worm gear **1834** and pinion gear **1838** together. The ball bearings **2048** may be made of a metallic material, such as stainless steel, to resist the sheer loading applied to the ball bearings **2048**. The worm screw **1814**, worm gear **1834**, pinion gear **1838**, spring **2040**, and release shaft **2030** may also be made of one or more metallic materials, such as stainless steel, and/or high-strength plastic materials.

(144) With reference to FIG. **63**, a retractor **2200** is provided that is similar in many respects to the retractors discussed above. The retractor **2200** includes a frame **2202** extending about a central opening **2204** and having operating mechanisms **2206** for connecting tissue engaging members to the frame **2202**. The operating mechanisms **2206** each include a slider **2208** and a slider linear drive **2210** for shifting the slider **2208** in directions **2212**, **2214**. The retractor **2200** also includes a clutch **2216** associated with the slider linear drive **2210**. The slider linear drive **2210** and the clutch **2216** are similar to the slider linear drive **1812** and clutch **1830** discussed above.

(145) Regarding FIG. **64**, the sliders **2208** may be made of a carbon fiber material which is radiolucent and provides less obstructed x-ray imaging of a surgical site during a surgical procedure. Each slider **2208** includes an arm, such as an inner member **2220**, and a base, such as an outer member **2222**, that are pivotally connected by a pivot pin **2224**.

(146) With reference to FIGS. **65** and **66**, the outer member **2222** has an upstanding wall **2226** that has a hole **2228** which receives the pivot pin **2224**. The inner member **2220** includes a pocket **2230** that receives the wall **2226**. The pocket **2230** has walls **2232** on opposite sides of the wall **2226** therebetween and guide the inner member **2220** as it pivots relative to the outer member **2222**. Further, the outer member **2222** may have peripheral walls **2240** that further guide the inner member **2220** as the inner member **2220** pivots relative to the outer member **2222**. Because the inner member **2220** and the outer member **2222** may be made of carbon fiber, the walls **2226**, **2240**, and **2232** are sized to provide sufficient rigidity to the material of the inner member **2220** and the outer member **2222** to resist fracture of the carbon fiber material.

(147) Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above described embodiments without departing from the spirit and scope of the invention, and that such modifications, alterations, and combinations, are to be viewed as being within the scope of the invention.

Claims

1. A retractor configured to enlarge an incision, the retractor comprising: a frame; a plurality of tissue engaging members configured for insertion into an incision; a plurality of sliders configured

to connect the tissue engaging members to the frame, each slider being shiftable between an extended position and a retracted position, wherein each slider is selectively shiftable between the extended position and the retracted position; and a worm screw engaged with a worm gear and rotatable in a first direction to cause the worm gear to rotate in a retracting direction, the worm screw being rotatable in an opposite, second direction to cause the worm gear to rotate a drive member associated with one of the sliders in an extending direction.

2. The retractor of claim 1, wherein the worm gear is operatively coupled to the drive member, and wherein the worm screw is configured to be self-locking to inhibit unintended rotation of the drive member and lock at least one of the plurality of sliders in position relative to the frame.

3. The retractor of claim 1, wherein the worm gear is operatively couplable to the drive member and rotatable about a first axis, and wherein the worm screw is rotatable about a second axis perpendicular to the first axis.

4. The retractor of claim 1, wherein each of the plurality of sliders includes a second slider, the retractor further comprising: a second drive member associated with the second slider; a second worm gear operably coupled to the second drive member; and a second worm screw engaged with the second worm gear and rotatable in a third direction to rotate the second worm gear along with the drive member in a second retracting direction transverse to the retracting direction of the second slider, the second worm screw being rotatable in an opposite, fourth direction to rotate the second worm gear along with the second drive member in a second extending direction transverse to the extending direction of the second slider.

5. The retractor of claim 1, wherein one slider of the plurality includes a slider body slidably connected to the frame and a support pivotally connected to the slider body, the support having a tissue engaging member portion connectable to one of the tissue engaging members.

6. The retractor of claim 1, wherein one slider of the plurality includes: a pivotal support having a tissue engaging member portion configured to be connected to one of the tissue engaging members and an actuator portion spaced from the tissue engaging member portion; and a rotatable actuator connected to the actuator portion of the support and having a tool-receiving portion configured to receive a separate tool for rotating the actuator and causing the support and the one tissue engaging member connected thereto to pivot relative to the frame.

7. The retractor of claim 6, wherein the pivotal support is pivotal about a pivot axis intermediate the tissue engaging member and the actuator portion along the support.

8. The retractor of claim 7, wherein the pivot axis is closer to the tissue engaging member portion than the actuator portion along the support.

9. The retractor of claim 8, wherein the rotatable actuator and the slider body include a ball and socket that permits the rotatable actuator to pivot relative to the slider body with pivoting of the support and the tissue engaging member connected thereto.

10. The retractor of claim 7, wherein one slider of the plurality includes a slider body slidably connected to the frame and the support is pivotally connected to the slider body.

11. A retractor configured to enlarge an incision, the retractor comprising: a frame; a plurality of tissue engaging members configured for insertion into an incision; a plurality of operating mechanisms configured to move the tissue engaging members apart to enlarge the incision; a slider of one of the operating mechanisms connectable to one of the tissue engaging members, the slider being connected to the frame and shiftable between an extended position and a retracted position; a linear drive mechanism associated with the slider and including a drive member configured to cause the slider to selectively shift between the extended and retracted positions; a worm gear operatively coupled to the drive member; a tool-receiving member of the linear drive mechanism operably coupled to the drive member; and a clutch of the linear drive mechanism configured to permit the tool receiving member to turn relative to the drive member, wherein the clutch is selectively disposable between a drive mode and a release mode.

12. The retractor of claim 11, wherein the clutch comprises a release shaft cooperatively configured

with a release actuator to couple and decouple the worm gear from the drive member to dispose the clutch respectively in the drive mode and the release mode.

13. The retractor of claim 11, wherein the release shaft comprises a head operatively configured with at least one coupling member of the drive member.

14. The retractor of claim 13, wherein the at least one coupling member comprises a plurality of ball bearings disposed into pockets of the worm gear; the plurality of ball bearings disposable into a radially outward position extending into the pockets of the worm gear and a radially inward position wherein the plurality of ball bearings are spaced apart from the worm gear decoupling the drive member.

15. A retractor configured to enlarge an incision, the retractor comprising: a frame; a plurality of tissue engaging members configured for insertion into an incision; a plurality of operating mechanisms configured to move the tissue engaging members apart to enlarge the incision; a slider of one of the operating mechanisms connectable to one of the tissue engaging members, the slider being connected to the frame and selectively shiftable between extended and retracted positions; a linear drive mechanism associated with the slider and including a drive member configured to cause the slider to shift to the retracted position; a worm gear operatively coupled to the drive member; a tool-receiving member of the linear drive mechanism operably coupled to the drive member; and a clutch of the linear drive mechanism configured to permit the tool receiving member to turn relative to the drive member, wherein the clutch is selectively disposable between a drive mode and a release mode, wherein the clutch comprises a release shaft comprising a head operatively configured with at least one coupling member of the drive member, and wherein the release shaft is operatively configured with a release actuator to couple and decouple the worm gear from the drive member to dispose the clutch respectively in the drive mode and the release mode.
